

MATHEMATIC CURRICULUM

THEME: NUMBERS AND NUMERATION
SUB-THEME: WHOLE NUMBERS

PRIMARY FOUR

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
1. Whole numbers	Pupils should be able to: 1. count in thousands up to one million 2. solve problems on quantitative reasoning	i. Counting up to one million in ones, tens, hundreds and thousands ii. Quantitative reasoning	1. Guides pupils to count by using bundles of sticks, seeds, etc in counting in tens and hundreds. 2. Guides pupils to build place value abacus and asks them to label columns. 3. Assists pupils in making overlay cards to be used in forming and writing numbers as shown below 5 0 0 0 0 3 0 0 0 6 0 0 7 0 4. Demonstrates the use of overlay cards. 5. Guides the pupils in activities to build numerals up to 99,999 with abacus. 6. Guides pupils to form and write 5-digit number using overlay cards. 7. Guides pupils to solve problems in quantitative aptitude e.g. complete the following diagram for an overlay card for the number. 5 3 6 0 5 0 0 0 3 0 0 □ 0 8. Designs series of activities to lead pupils to count meaningfully up to one million	1. Count using their counters e.g. sticks and match sticks in tens and hundreds. 2. Label columns of place value abacus. 3. Make overlay cards. 4. Observe and demonstrate the use of overlay cards. 5. Count numbers with abacus. 6. Make use of overlay cards to write 5-digit numbers. 7. Solve problems on quantitative aptitude.	Counters: Sticks Match sticks etc Abacus, Overlay card, Cardboard paper.	Pupils to: 1. count numbers up to 999,999 and one million; 2. solve problems on counting numbers up to one million; 3. solve Problems in quantitative reasoning involving whole numbers.

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TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
Whole numbers contd.	Pupils should be able to: 3. apply knowledge of counting to local counting in groups of: ▪ fives (yams, oranges, onions etc) ▪ market days ▪ sevens (weeks and days) ▪ 60's (minutes and seconds etc) 4. solve problems on quantitative reasoning involving whole numbers 5. state the place value of a digit in four digit numbers	iii. Counting in 5's, 7's and 60's etc. iv. Quantitative reasoning v. Place values: thousands, hundreds, tens and units Th, H, T, U	1. Organizes a short visit to the nearby market for pupils to observe grouping of commodities for sale e.g. oranges, yams, onions. 2. Guides pupils to count in 5's using the local common activities. 3. Guides pupils to count in 7's using days of the week. 5. Makes use of clock faces to lead pupils count in 60's. 6. Organizes activities to make pupils change days to weeks, seconds to minute and also minutes to hours, vice versa. 7. Fill up the gap below: 100 200 300 ? 8. Guides pupils to state the place value of a digit using abacus. 9. Writes the place value of a 4-digit number as Th, H, T, U and explains that: Th = Thousands H = Hundreds T = Tens U = Units 10. Guides pupils to write the place value of each digit in the number 6435 as: 5 = units 3 = 3tens 4 = 4 hundreds 6= thousands 11. Writes 6 4 3 5 on the board and asks pupils to write the value of each digit of the number e.g., 5 = 5 3 = 30 4 = 400 6 = 6000	1. Observe the grouping of local commodities in the market. 2. Count in 5's e.g. 5, 10, 15, 20, 25. 3. Count in 7's e.g. 7, 14, 21, 28. 4. Count in 60's e.g. 60, 120, 180, 240. 5. Change days to weeks e.g. 21days = 3 weeks, 26days = 3weeks 5days 6. Change minute to hours e.g. 260mins = 4 hours 20mins 7. Change seconds to minutes e.g. 180 seconds s = 3 minutes 8. Solve some problems on quantitative aptitude involving whole numbers. 9. Write the place value of the digits in a four digit number. 10. Write the place value of each digit in a given number such as the number written on the board as: 5 = 5 3 = 30 4 = 400 6 = 6000 11. Solve some problems on the place value involving four digit numbers.	Counters, available objects in the locality, clock faces, calendar, etc. Abacus, Spike, and Thread, beads	Pupils to: 4. Solve problems in groups of five, seven and sixty; 5. write the place value of each digit in a given 4-digit number: (a) using ▪ spike abacus ▪ pocket abacus ▪ thread abacus ▪ thread Beads (b) without the use of abacus.

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Whole numbers contd.	Pupils should be able to: 6. count Roman numerals up to 100 (I to C); 7. solve problems on quantitative reasoning involving use of roman numeral; 8. order whole numbers up to 1000 using the symbol < and >; 9. solve problems on quantitative reasoning involving ordering of whole numbers.	vi. Roman numerals (I to C) I for one V for five X for ten L for fifty C for hundred IX for nine XX for twenty XXX for thirty vii. Quantitative reasoning: XL for forty LXX for seventy LXXX for eighty viii. Ordering of whole numbers up to 1000. ix. Quantitative reasoning.	1. Guides pupils to use fingers to represent numbers in roman numerals I – X (1 to 10) 2. Guides pupil to form Roman numeral: - I (one finger), - V (five fingers) - X (ten fingers) 3. Uses Roman numeral chart to lead pupils to read and write the Roman numerals I to C 4. Gives examples on quantitative aptitude. e.g. find the missing number below. L ↑ 50 X ↑ 10 VI ↑ ? 5. Guides pupils in ordering of numbers 1 – 500 on numeral chart. 6. Leads pupils to recall the meaning of the symbol (< and >) 7. Guides pupils to order two numbers e.g. 546 and 174 as 546 > 174 or 174 < 546 8. Gives more exercises on ordering of whole numbers, 9. Guides pupils to solve some quantitative aptitude problems involving ordering of whole numbers.	1. Study the Roman numerical chart and flash cards 2. Practice the number formation system for Roman numerical (I x C) using fingers. 4. Read and write Roman numerals up to 100 and also write the value of each numerals in the Hindu/Arabic form 5. Write all the Roman numerals I to C in their exercise book 6. Solve examples on quantitative aptitude involving roman numerals 7. Revises order of numbers 1-500 using number chart 8. Recall the meaning of the symbol (< and >) 9. Order any two given numbers 10. Solve problems on ordering whole numbers 11. Solve problems on quantitative aptitude involving ordering of whole numbers	Roman numerical Charts Flash cards. Roman numeral chart Flash cards. Number charts Flash cards	Pupils to: 6. read given Roman numerals and state the corresponding values in the Hindu/Arabic Form; 7. write the value of the Roman numeral I - C in the Hindu/Arabic form; 8. write 1 – 100 in roman numerals; 9. solve problems on quantitative aptitude; 10. order given whole numbers; 11. solve more problems on quantitative aptitude.

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2. LCM	Pupils should be able to find the LCM of numbers up to 9	i. LCM of two numbers up to 9	1. guides pupils to find LCM of one digit number	1. find LCM by use of factor method.	cardboard	Pupils to: 1. obtain LCM of two numbers by identifying the lowest of the common multiples; 2. find the LCM by factor method.
3. HCF	Pupils should be able to find the HCF of 2-digit numbers.	ii. HCF of 2-digit numbers.	1. Guides pupils to use flannel board and flash cards to identify HCF numbers. 2. Guides pupils to identify HCF of two digit numbers.	1. Use flannel board and flash cards to identify common factors of given numbers. 2. Find the HCF of two digit numbers.	Flannel board, Flash card, Number cards,	Pupils to find the common factors of given numbers and hence determine the HCF.

MATHEMATIC CURRICULUM

PRIMARY FOUR

THEME: NUMBERS AND NUMERATION
SUB-THEME: FRACTION NUMBERS

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
1. Fraction	Pupils should be able to: 1. differentiate between proper and improper fractions; 2. change improper fractions to mixed numbers and vice versa; 3. apply fractions in sharing commodities in home, market, school, etc; 4. solve quantitative reasoning problems involving fractions; 5. use decimal fraction up to hundredths; 6. solve quantitative. Reasoning problems in decimal fractions; 7. obtain equivalent fractions of a given fraction; 8. ordering pairs of fractions; 9. solve quantitative reasoning on equivalent fractions.	i. Proper and improper fractions and mixed numbers ii Mixed number iii Quantitative reasoning iii. Decimal fraction up to tenths and hundredth iv. Quantitative reasoning v. Equivalent fractions vi ordering fractions vii Quantitative reasoning	1. Guides pupils to divide an objects e.g. an orange into two equal parts to show a proper fraction 2. Guides pupils to identify what is proper and improper fraction as well as a mixed number. 3. Guides pupils to Convert from improper fractions to mixed numbers and vice versa 4. Guides pupils to apply fractions in sharing commodities in homes market Schools etc. 5. Guides pupils to identify decimal numbers in decimal fraction: tenths hundredths as fractions with 10 and 100 as denominators respectively. 6. Solve problems in quantitative reasoning. 7. Guides pupils to state three or four equivalent fractions of a given fraction. 8. Guides pupils to notice that when fractions have the same denominator the fraction with smaller numerator is less e.g. $\frac{6}{10} \text{ and } \frac{9}{10} \text{ implies that}$ $\text{hence } \frac{3}{5} < \frac{9}{10}$ 9. Guides pupils to solve quantitative aptitude problems on equivalent fractions.	1. Explains the meaning of proper and improper fractions and mixed fractions. 2. Cut objects e.g. orange into equal parts to show a proper fraction 3. Convert improper fractions to mixed numbers and vice versa. 4. Apply fractions to share commodities in homes market schools etc. 5. Explain decimals in 10's , 100's as fractions with 10 and 100 as denominators 6. Solve problems in quantitative reasoning. 7. Give equivalent fractions of given fractions. 8. Order a pair of fraction by comparing their new numerators. 9. Solve some quantitative aptitude problems involving equivalent fractions	Orange, Paper cuttings of different shapes Fraction charts Oranges, Squares, Cardboards. Flow chart of quantitative reasoning Flash Cards	Pupils to: 1. classify a given set of fractions into proper and improper fractions; 2. convert improper fraction to mixed number and vice versa; 3. solve more conversion problems on fractions; 4. write decimal fractions with denominators ten and hundred; 5. solve problems in quantitative reasoning; 6. find the equivalence of given fractions; 7. order given pairs of fractions using < or >; 8. solve problems on quantitative aptitude reasoning involving equivalent fractions.

MATHEMATIC CURRICULUM

THEME: BASIC OPERATIONS
SUB-THEME: BASIC OPERATIONS

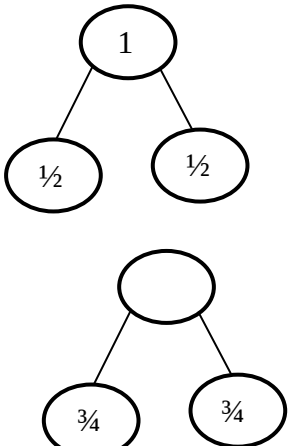
PRIMARY FOUR

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	PUPILS		
1. Addition and subtraction	Pupils should be able to: 1. add and subtract whole numbers in Th. H. T. U with or without renaming; 2. carryout correct addition and subtraction in everyday life activities; 3. add and subtract three 4-digit numbers taking two at a time; 4. solve quantitative aptitude problems involving addition and subtraction of whole numbers of four digit;	i. Addition and subtraction of whole numbers in Th. H. T. U With or without renaming. ii. Addition and subtraction of three 4-digit numbers taking two at a time. iii. Quantitative Reasoning. iii. Quantitative Reasoning on addition and subtraction of whole numbers of four digits.	1. Guides pupils to use abacus with Th. H.T. U for addition and subtraction of three 4-digit numbers with or without renaming. 2. Guides the pupils to note the importance of place value in performing correct addition and subtraction in everyday activities. 3. Guides pupils to carry out the correct addition and subtraction. 4. Guides pupils through addition and subtraction of three 4-digit numbers taking two at a time e.g. $1816 + 2672 + 4297$ 5. Guides pupils to solve quantitative aptitude problems involving addition and substation of three 4-digit numbers e.g. $ \begin{array}{r} 4600 \quad \text{—————} \quad 9200 \\ \quad \quad \quad \diagup \quad \quad \quad \diagdown \\ \quad \quad \quad \text{—————} \quad \quad \quad \text{—————} \\ \quad \quad \quad \quad \quad \quad \quad \quad 13800 \end{array} $	1. Use abacus to subtract three 4-digit numbers. 2. Apply place value in addition and subtraction and need for correct addition and subtraction. 3. Solve problems that require addition and subtraction in at least four digit numbers. 4. Add and subtract three 4-digit numbers taking two at a time e.g. $1816 + 2672 + 4297$ 2. $4297 - 2672 - 1816$ 5. Solve quantitative aptitude problems involving additions and subtraction of three 4-digit numbers	Abacus and charts	Pupils to: 1. add and subtract three 4-digit numbers with or without renaming; 2. add and subtract three 4-digit numbers taking two at a time; 3. solve quantitative aptitude problems involving additions and subtraction of three 4-digit numbers taking two at a time.

MATHEMATIC CURRICULUM

THEME: BASIC OPERATIONS
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PRIMARY FOUR

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	PUPILS		
Addition and subtraction contd.	Pupils should be able to: 5. add and subtract: ▪ two proper fractions ▪ improper fractions and mixed fractions 6. correctly add and subtract proper and improper fractions in everyday life activities; 7. solves quantitative aptitude problems involving addition and subtraction of fractions.	iii. Addition and subtraction of ▪ two proper fractions ▪ improper fractions and mixed numbers. iv. Quantitative reasoning on addition and subtraction of fractions.	1. Guides the pupils to revise addition of fractions e.g. $\frac{1}{2} + \frac{3}{4} = \frac{2}{4} + \frac{3}{4} = \frac{5}{4} = 1\frac{1}{4}$ 2. Guides pupils to add and subtract proper and mixed numbers. 3. Guides pupils to give examples of everyday activities where addition and subtraction of fractions and decimals are used. 4. Guides pupils to solve quantitative aptitude problems involving addition and subtraction of fractions e.g. 	1. Revise addition and subtraction of fractions. 2. Carry out addition and subtraction involving proper fraction and mixed numbers. 3. Give examples of everyday life activities where accuracy of addition and subtraction of fractions and decimals are required. 4. Solve quantitative aptitude problems involving addition and subtraction of fractions.	Fraction Charts Quantitative aptitude Chart	Pupils to: 4. add and subtract proper fractions and mixed fractions; 5. give three or more examples of everyday life activities that require addition and subtraction of fractions; 6. solve quantitative aptitude problems involving addition and subtraction of fractions.

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			TEACHER	PUPILS		
Addition and subtraction contd.	Pupils should be able to: 8 add and subtract decimal up to 3 places; 9. correctly add and subtract decimal; 10. solve problems on quantitative reasoning involving addition and subtraction of decimals.	v. Addition and subtraction of decimals up to 3 decimal numbers. vi. Quantitative reasoning on decimal numbers	1. Guides pupils to solve problems on addition and subtraction of decimals to 3 decimal places e.g. $4.263 + 5.123 = 9.386$ 2. Leads pupils to correctly add and subtract decimals in everyday life activities. 3. Guides pupils to solve quantitative aptitude problems involving addition and subtraction of decimals e.g. $\begin{array}{r} 18.22\text{cm} \\ + 11.45\text{cm} \\ \hline 29.67\text{cm} \end{array}$ $\begin{array}{r} 14.28\text{kg} \\ + 9.70\text{kg} \\ \hline \boxed{?} \end{array}$	1. Add and subtract series of problems involving decimal numbers up to 3 decimal places. 2. Correctly add and subtract decimal numbers giving examples 3. Apply addition and subtraction to every life activities 4. Solves quantitative aptitude problems involving addition and subtraction of decimals numbers with up to 3 decimal places.	Charts addition cards Quantitative aptitude charts containing worked problems	Pupils to: 7. give four examples of everyday activities that require correct addition and subtraction of decimal numbers; 8. solve problems on addition and subtraction on decimal numbers up to 3 decimal places; 9. solve quantitative aptitudes problems involving addition and subtraction of decimal numbers with up to 3 decimal places.

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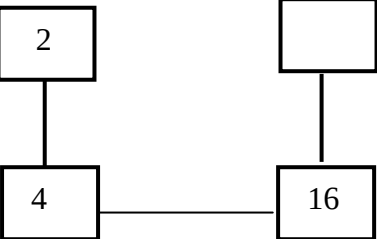
THEME: BASIC OPERATIONS
SUB-THEME: BASIC OPERATIONS

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	PUPILS		
2. Multiplication	Pupils should be able to: 1. multiply whole numbers by 2-digit number not exceeding 50; 2. solves quantitative aptitude problems involving multiplication of whole numbers by 2-digit numbers; 3. multiply decimals numbers by 2-digit numbers; 4. solve quantitative aptitude problems involving multiplication of decimal numbers by 2 digit numbers.	i. Multiplication of whole numbers by 2-digit numbers not exceeding 50. iii. Quantitative reasoning on multiplication of whole numbers iii. Multiplication of decimals by 2-digit numbers. iv. Quantitative reasoning on multiplication of decimal numbers.	1. Guides pupils to Revise multiplication of whole numbers. 2. Guides pupils to carry out multiplication of whole numbers in 2-digit numbers e.g. 40×12 . 3. Guide pupils to solving quantitative aptitude problems on multiplication of whole numbers with 2-digit numbers e.g. 150 — 12 — 1800 200 — — 2000 ? — 10 — 3,500 4. Guides pupils to solve multiplication of decimal numbers by 2-digit numbers. 5. Guides pupils to solve quantitative aptitude problems involving multiplication of decimal numbers by 2 digit whole numbers	1. Carry out multiplication of whole numbers by 2-digit numbers. 2. Solve problems involving multiplication of whole numbers by 2-digit numbers. 3. Solves quantitative aptitude problems involving multiplication of whole numbers by 2-digit numbers not exceeding 50 1. Give examples of everyday activities that require correct multiplication of decimals with 2-digit numbers. 2. Solve problems on multiplication of decimal number with 2 digit number. 3. Solve series of quantitative aptitude problems involving multiplication of decimal numbers 2 digit whole numbers	Flash Cards Multiplication charts, etc Quantitative aptitude Chart, etc	Pupils to: 1. multiply whole numbers by 2-digit numbers not exceeding 50; 3. solve quantitative aptitude problems involving multiplication of whole numbers by 2-digit numbers not exceeding 50; 4. multiply decimals by 2-digit numbers; 5. solve quantitative aptitude problems involving multiplication of decimal numbers by 2 digit numbers.

MATHEMATIC CURRICULUM

THEME: BASIC OPERATIONS
SUB-THEME: BASIC OPERATIONS

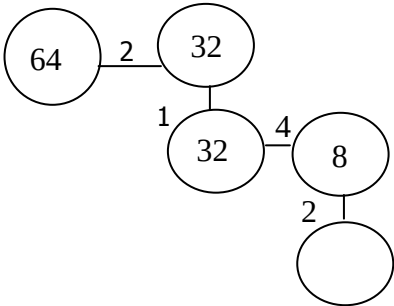
PRIMARY FOUR

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	PUPILS		
Multiplication contd.	Pupils should be able to: 5. calculate the squares of 1 and 2 digit numbers; 6. identify objects with perfect faces like cubes and Square shapes; 7. solve quantitative aptitude problems involving squares; 8. find square roots of perfect squares up to 400; 9. solve quantitative aptitude problems of square roots of perfect squares not greater than 400.	v. Squares of 1-digit and 2-digit numbers. vi. Quantitative reasoning on squares of numbers. vii. Square root of perfect squares up to 400 using: - square chart - factor method. viii. Quantitative reasoning on square root of numbers not greater than four hundred.	1. Guides pupils to find the square of a number say 3 by locating the intersection of 3 by 3 on the vertical and horizontal rows and columns respectively on the square chart to get 9. 2. Guides pupils to identify objects of perfect square faces. 3. Guides pupils to solve quantitative aptitude problems involving square of 1-digit and 2-digit numbers e.g.  3. Guides pupils in locating the square root of any perfect square on the chart e.g. square root 256 is 16. 4. Guides pupils to find square roots by factor method e.g. $256 = 4 \times 4 \times 4 \times 4$ 5. Guides pupils to solve quantitative aptitude problems involving Square root of perfect squares not greater than 400, illustrating with charts e.g. 	1. Use square chart to find squares of any given number. 2. Give examples of objects with perfect square faces. 3. Solve series of quantitative aptitude problems involving squares of 1- digit and 2-digit numbers. 4. Find square root of perfect squares not exceeding 400 using the square root chart. 5. Find square root of perfect squares by factor Methods. 6. Solve quantitative aptitude involving square root of numbers not exceeding 400.	20 by 20 square chart Charts on quantitative reasoning Square root chart charts	Pupils to: 7. find the square of given number using 20 by 20 square chart; 8. give example of objects with perfect square faces in their homes and school; 9. solve quantitative aptitude problem involving finding of squares; 10. find square root of given numbers using 20 by 20 square chart; 11. find square root of perfect squares using factor method; 13. solve quantitative aptitude problems involving squares and square root of numbers.

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THEME: BASIC OPERATIONS
SUB-THEME: BASIC OPERATIONS

PRIMARY FOUR

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	PUPILS		
3. Division	Pupils should be able to: - divide two or three digit numbers by: - numbers up to 9 with or without remainder - multiples of 10 up to 50 - Solve quantitative aptitude problems involving division.	- Division of two or three digit numbers by: - numbers up to 9 with or without remainder - multiples of 10 up to 50 - Quantitative reasoning on division.	- Guides pupils to divide 2 or 3 decimal by numbers not more than 9. - Guides pupils to divide a given number by 10, 20, 30 etc. - Guides pupils to solve quantitative aptitude problems such as: 	- Divide 2 and 9 digit numbers by numbers not more than 9. - Divide given numbers by 10,20,30 etc. - Solve quantitative aptitude problems involving division.	Addition charts charts	Pupils to: - divide a given two or three digit number by numbers between 2 and 9; - divide a given two or three digit number by 10 up to 50; - solve quantitative aptitude problems involving division.

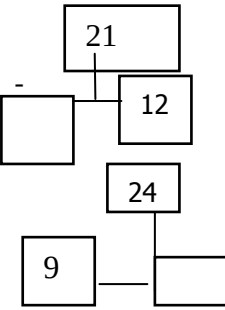
THEME: BASIC OPERATIONS
SUB-THEME: DERIVED FUNCTION

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	PUPILS		
Estimate	Pupils should be able to give meaningful estimate of sums and products of numbers.	i. Estimate sums and products. ii. Quantitative reasoning problem involving number line.	Guides pupils to find estimate of sum and product of numbers	Estimate sums and products numbers.	Charts on estimates	Pupils to: 1. estimate the sum and product of numbers, height, lengths, and widths within and outside the school; 2. carry out estimate involving sum and product of numbers.

MATHEMATIC CURRICULUM

PRIMARY FOUR

THEME: ALGEBRAIC PROCESSES
SUB-THEME: ALGEBRAIC OPERATIONS

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
1. Open sentence	Pupils should be able to : 1. define open sentences; 2. find missing number in an open sentences; 3. solve related, quantitative aptitude problems on open sentences involving multiplication and division.	1. Open sentences. 2. Quantitative reasoning on open sentences	1. Guides pupils to find missing numbers in example: 1. $9 + \square = 15$ 2. $18 - \square = 7$ 2. Guides pupils to solve quantitative aptitude problems of the form 	1. Solve series of problems involving open sentences. 2. Solve related quantitative aptitude problems on open sentences.	1. Chart containing worked examples on open sentences	Pupils to: 1. solve given problems on open sentences involving the four basic operations; 2. find missing numbers in quantitative aptitude diagrams drawn from open sentences; 3. solve some problems of quantitative aptitude involving open sentences.

MATHEMATIC CURRICULUM

THEME: MENSURATION AND GEOMETRY SUB-THEME: PRIMARY MEASURE

PRIMARY FOUR

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	PUPILS		
1. Money	Pupils should be able to: 1. solve problems on addition of money; 2. solve quantitative aptitude problems on additions of money; 3. solve problem on subtraction of money; 4. solve quantitative aptitude problems involving subtraction of money.	i. Addition of money ii. Quantitative reasoning involving addition of money iii. subtraction of money iv. Quantitative reasoning involving subtraction of money	1. Guides pupils to solve problems on addition of money e.g. $\text{N}7.50 + \text{N}6.30 = \text{N}15.80$ $\text{N}3.80 + \text{N}7.30 = \text{N}11.10$ 2. Guides pupils to solve quantitative aptitude problems involving addition of money. 3. Guides pupils to relate correctly, addition of money in business transactions as a customer or dealer of goods and services. 4. Guides the pupils to Engage in mock buying and selling which will involve the collection of change. 5. Guides pupils on the subtraction of money e.g. $\text{N}9.50 - \text{N}6.20 = \text{N}3.30$ 6. Leads pupils to note the importance of correct subtraction of money in business transaction. 7. Solves quantitative aptitude problems involving subtraction of money.	1. Solve some problems on addition of money. 2. Solve quantitative aptitude problems involving addition of money. 3. Relate on correct addition of money in business transaction as a customer or dealer of goods and services. 4. Act as shop keepers or customers in mock buying and selling which will involve the collection of change. 5. Solve problems on the subtraction of money. 6. State correctly the importance of correct subtraction of money in business transaction 7. Solve some quantitative aptitude problems related to money subtraction.	1. Empty containers of: Milk, Geisha, Tomato, Sugar etc. 2. Basic addition table for money. 3. Real and model money.	Pupils to: 1. solve given problems on addition of money; 2. solve quantitative reasoning problems involving money; 3. state 2 or three reasons why addition of money is important in business transactions; 4. solve quantitative aptitude problems involving the subtraction of money; 5. solve quantitative problems.

MATHEMATIC CURRICULUM

THEME: MENSURATION AND GEOMETRY SUB-THEME: PRIMARY MEASURE

PRIMARY FOUR

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	PUPILS		
Money contd.	Pupils should be able to: 5. multiply money by a whole number; 6. solve some quantitative reasoning problems on multiplications of money; 7. divide money by a whole number; 8. solve quantitative reasoning problems in real life situations; 9. calculate profit and loss; 10. solve quantitative aptitude problems on profit and loss.	iv. Multiplication of money by a whole number. v. Division of money by a whole number. vi. Calculation of profit. vii. Quantitative reasoning on profit and loss.	1. Leads pupils on multiplication of money by whole number using sum cards e.g. ₦7 x 3 = ₦21. 2. Solves quantitative aptitude problems related to multiplications of money. 3. Guides pupils to solve problems on division of money by a whole number. 4. Guides pupils to solve quantitative aptitude problems on division of money. 5. Gives exercise on division of money by a whole to pupils. 6. Sets up a shopping corner in the classroom and engage pupils in buying and selling. 7. Leads pupils to realize that when one sells an article at a price higher than the cost price then a profit is said to have been made. 8. Guides pupils to calculate the profit by subtracting cost price (CP) from the selling price (SP) i.e. $\text{profit} = \text{selling price} - \text{cost price}$ $\text{profit} = \text{SP} - \text{CP}$ 9. Guides pupils to solve problems on quantitative reasoning involving profit and loss	1. Solves problems on multiplication of money by whole number. 2. Solve quantitative aptitude problems related to multiplication of money. 3. Solve problems on division of money by a whole number. 4. Solve quantitative aptitude problems on division of money. 5. Calculate profit and loss. 6. Solve problems on profit and loss. 7. Solve problems on quantitative reasoning involving profit and loss. 8. State skills in making profit in real life through entrepreneurship and enterprise activities. 9. Solve problems on quantitative reasoning involving profit and loss.	1. Cards on multiplication of money by whole numbers. 2. Real and model of money. 3. Empty packets of: Sugar, Match, Empty tins of: Bournvita, Tomatoes, milk etc. Real and models of money.	Pupils to: 6. Solve problems on multiplication of money by whole numbers. 7. Solve problems on division of money by whole numbers. 8. Solve quantitative aptitude problems involving division of money. 9. Solve problems on the calculations of profit and loss. 10. Solve problems on quantitative reasoning involving profit and loss. 11. State ways on how to make profit in life.

MATHEMATIC CURRICULUM

THEME: MENSURATION AND GEOMETRY
SUB-THEME: PRIMARY MEASURE

PRIMARY FOUR

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	PUPILS		
2. Length	Pupils should be able to: 1. estimate distances in kilometers and length in meters or centimeters and compare with measurements; 2. add and subtract length.	i. Estimating lengths and Measurements. ii. Addition and subtraction of lengths	1. Guides pupils to estimates and measure small objects in centimeters 2. Guides pupils to estimate and measure bigger objects in meters and centimeters and record all their results in a tabular form. 3. Guides pupils to compare their results. 4. Guides pupils to add and subtract lengths e.g. $8\text{km } 120\text{m} + 4\text{km } 4\text{m} = 12\text{km } 124\text{m}$ $6\text{km} + 3\text{km } 8\text{m} = 9\text{km } 8\text{m}$ $7\text{km } 45\text{m} - 5\text{km } 20\text{m} = 2\text{km } 25\text{m}$	1. Estimate and measure small objects in centimeters bigger objects in meters and centimeters. 2. Make comparison of their results add and subtract given lengths.	Tape, Meter, rule, etc.	Pupils to: 1. estimate given length and distances 2. estimate given objects and distances; 3. compare estimates and actual measurement; 4. solve given problems on addition and subtraction of lengths.
3. Weight	Pupils should be able to: 1. compute addition and subtraction of weights using Kg and g; 2. multiply and divide weight by whole numbers.	i. Addition and subtraction of weight in kilograms (kg) and grams (g). ii. Multiplication and division of weights in (kg) and (g) by whole numbers.	1. Guides pupils to Revise conversion : $1\text{kg} = 1000\text{g}$ and leads pupils to solve problems in addition and subtraction of weights $412\text{g} + 319\text{g} = 731\text{g}$ $2\text{kg } 457\text{g} + 3\text{kg } 716\text{g} + 412\text{g} = 6\text{kg } 585\text{g}$ $15\text{kg } 437\text{g} - 4\text{kg } 39\text{g} = 11\text{kg } 398\text{g}$ 2. Leads pupils to solve problems on multiplication and division of weights in (kg) and (g) by whole numbers $212\text{g} \times 3$ $2\text{kg } 212\text{g} \times 9$ $216\text{g} \div 3$ $3\text{kg } 560 \div 7$.	1. Solve the given problems on addition and subtraction of weights. 2. Solve problems on multiplication and division of weights by whole numbers.	Weighing scale, Scale balance, etc.	Pupils to: 1. solve problems on addition and subtraction of weights; 2. multiply and divide weights by whole numbers.

MATHEMATIC CURRICULUM

THEME: MENSURATION AND GEOMETRY SUB-THEME: PRIMARY MEASURE

PRIMARY FOUR

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	PUPILS		
4. Time	Pupils should be able to: 1. give time on the clock, read calendar and write dates; 2. solve problems on quantitative aptitude problems related to time; 4. use the notation am and pm for time of the day.	i. Time on the clock, calendar and dates. ii. Quantitative reasoning on time. iii. Use of the notation am and pm for time of the day	1. Guides pupils to indicate important times of the school day such as first lesson (8.10), break (11.10) and closing (1.30). 3. Guides pupils to revise telling of time on the clock face. 3. Leads pupils to tell their birth days which are recorded on the board using the format: 23/06/1998 (day/Moth/Year). 4. Guides pupils to solve problems on quantitative reasoning related to time. 5. Leads pupils to explains using the clock to indicate the meaning of am (ante meridian – before noon) and pm (post meridian – after noon) e.g. time for first lesson is 8.10am and time for closing is 1.30pm.	1. Indicate important times of the school day on a clock. 2. Indicate their birthdays and other important dates using the format on the board. 3. Solve problems on quantitative aptitude related to time. 4. Indicate times of the day that should be am and those that should be pm.	Clocks, Calendars and clock charts	Pupils to: 1. read given time on clock or charts. 2. indicate important activities at their homes and times when they take place; 3. write the dates of some given important events; 4. solve quantitative aptitude problems involving time; 5. record three important times of the (school) day and the activities associated with them indicating which times are am and which are pm.

MATHEMATIC CURRICULUM

PRIMARY FOUR

THEME: MENSURATION AND GEOMETRY
SUB-THEME: SECONDARY MEASURE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	PUPILS		
5. Area	Pupils should be able to: 1. find area of rectangles using the formula; 2. calculate areas of Shapes, farmland etc in the environment; 3. solve problems on quantitative aptitude related to areas.	i. Area of rectangles including squares ii. Areas of farmlands and towns or cities iii. Quantitative aptitude on area	1. Guides pupils to find area using square units. 2. Leads pupils to find area by using formula. 3. Guides pupils to find areas of shapes, farmlands e.t.c. in their environment. 4. Guides pupils to develop interest in find areas of shapes, structures and farmlands in their environment. 5. Guides them to solve problems on quantitative aptitude on areas.	1. Find small areas using unit squares. 2. Use formula to find areas of rectangles and squares. 3. Find areas of shapes, farmlands etc in their environment. 4. Solve problems on quantitative aptitude related to areas.	Cardboard and unit square chart; Classrooms, Desktops, Farmland etc	Pupils to: 1. find areas using formula; 2. find large areas in square meters and hectares.
6. Capacity	Pupils should be able to: 1. add and subtract in liters; 2. multiply and divide in liters with whole numbers; 3. solve problems on quantitative aptitude related to addition, subtraction, multiplication and division in involving liters.	i. Addition and subtraction involving liters. ii. Multiplication and division involving liters	1. Leads pupils to solve problems in addition and subtraction involving liters e.g. $7.5 \text{ liters} + 2.7 \text{ liters} = 10.2 \text{ litres.}$ 2. Leads pupils to solve problems on multiplication and division involving liters e.g. $9.2 \text{ liters} \times 6 = 18.12 \text{ litres}$ $2.16 \text{ liters} \div 6 = 0.36 \text{ litres}$ 3. Guides pupils to solve problems on quantitative reasoning involving multiplication and division using liters.	1. Solve problems in addition and subtraction involving litres. 2. Solve problems on multiplication and division involving liters. 2. Solve problems on quantitative aptitude involving litres.	1. Card board	Pupils to: 1. add and subtract given problems in liters. 2. divide and multiply with whole number in problems involving liters. 3. solve problems on quantitative aptitude on multiplication and division involving litres.

MATHEMATIC CURRICULUM

THEME: MENSURATION AND GEOMETRY SUB-THEME: SHAPES

PRIMARY FOUR

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	PUPILS		
1. Plane shapes	Pupils should be able to: 1. identify symmetrical plane shapes; 2. locate line(s) of symmetry of objects in the school and homes; 3. distinguish between horizontal and vertical lines; 4. Indicate the four cardinal points and relate the setting and rising of the sun on the cardinal points of the East and West.	i. Symmetry or non-symmetry of plane shapes: ▪ Square ▪ Rectangle ▪ Equilateral and Isosceles triangles ▪ Rhombus ▪ Trapezium ▪ Parallelogram ▪ kits iii. Horizontal and vertical lines iv. Cardinal points: • North • South • East • West	2. Guides pupils to fold the cut out plane shapes to find their line(s) of symmetry. 3. Leads pupils to locate the line(s) of symmetry of each of the plane shapes. 4. Leads pupils to appreciate symmetry in life situations. 5. Guides the pupils to locate vertical and horizontal lines in the classroom and other materials around them. 6. Guides the pupils to locate the four cardinal points and relate them to the geography of Nigeria. 7. Leads pupils to relate the rising and setting of the sun to the cardinal points.	1. Carryout the folding and cutting out plane shape to find the line(s) of symmetry. 2. Locate the line(s) of symmetry of the plane shape. 3. Locate various horizontal and vertical lines in the classroom and other materials around them. 4. Locate the four cardinal points and relate them to the geography of Nigeria. 5. Relate the rising and setting of the sun to the cardinal points.	Plane shapes, etc charts, plane shapes, etc Chart and model of the four cardinal points.	Pupils to: 1. Identify line(s) of symmetry of given plane shapes; 2. locate line of symmetry of given objects in their homes; 3. locate the horizontal and vertical lines on given objects; 4. mention five objects or materials that have horizontal and vertical lines in their environment; 8. locate given cardinal points on the chart and model; 9. mention the location of two given capitals of states in Nigeria.

MATHEMATIC CURRICULUM

THEME: MENSURATION AND GEOMETRY SUB-THEME: SHAPES

PRIMARY FOUR

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	PUPILS		
2. Three dimensional shapes	Pupils should be able to: 1. distinguish between open and closed shapes; 2. identify the uses of 3-dimensional shapes in homes and their environment.	v. Cuboids, cubes and cylinders.	1. Guides pupils to separate the open shapes and closed shapes and discuss their features. 2. Leads pupils to identify uses of 3-dimensional shapes in homes and environment.	1. Note and write down the features of open and closed shapes. 2. Identify the uses of 3-dimensional shapes in homes and environment.	Cartons, Milk tins, Plastic buckets, etc. Water tanks.	Pupils to distinguish between open and closed shapes from a given collection of shapes.

MATHEMATIC CURRICULUM

THEME: EVERYDAY STATISTICS

PRIMARY FOUR

SUB-THEME: DATA COLLECTION AND PRESENTATION

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
Bar Graph	Pupils should be able to: 1. draw bar graphs; 2. read bar graphs; 1. identify bar graph mode; 2. identify the most common events/ data in daily life activities	i. Bar graph ii. Bar Graph Mode	1. Guides pupils to Revise pictogram and idea of mode. 2. Guides pupils to Generate and presents data for the construction of bar graphs. 3. Leads pupils to read and interpret information in bar graphs. 4. Leads pupils to use bar graphs in presenting information. 5. Guides pupils to identify results involving bar graphs and mode of giving data. 6. Leads pupils to determine bar graph mode and least value from bar graphs. 7. Leads pupils to identify and find the mode of bar graphs of such statistical data as population distribution, amount of rainfall by months etc.	1. Construct bar graphs by representing given data on bar graphs. 2. Read and Interpret information from bar graphs 3. Use of bar graphics in representing information in their environment. 4. Identify the relation between bar graph and mode of a given data. 5. Give the mode on the bar graph and also the least value from bar graphs.	1. Cardboard showing horizontal and vertical bar graphs. 2. Mathematical game or demonstration designed by the teacher. 3. Bar Graph chart	Pupils to: 1. construct horizontal and vertical bar graphs of a given data; 2. solve problems from their environment where data can be represented using bar graphs; 3. identify bar graph; 4. determine the mode of a bar graph generated from their environment.

MATHEMATIC CURRICULUM

THEME: NUMBERS AND NUMERATION
SUB-THEME: WHOLE NUMBERS

PRIMARY FIVE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
1. Whole numbers	Pupils should be able to: 1. count in thousands and millions 2. apply counting of large numbers such as in population of states or country 3. solve quantitative aptitude problems related to thousand and millions 4. identify prime numbers less than 100.	i. Meaningful counting in thousands and millions. ii. Quantitative reasoning. iii. Identification of prime number less than 100.	1. Guides pupils to use abacus to form and read given number e.g 895,643. 2. Guides pupils to design various practices for counting and ordering numbers in thousands and millions. 3. Guides pupils to write several numbers in thousand and millions in words and figures. 4. Guides pupils to solve quantitative reasoning, problem on counting in thousand and millions e.g. 30,000 600 15,000 50 1. Guides pupils to find factors of numbers ranging from 1 to 100 in order, leading them to prepare a table of factor chart. 2. guides pupils to identify prime numbers in each set. 3. guides pupils to express numbers as product of prime factors.	1. Form and read numbers using a abacus. 2. Count and order numbers. 3. Read and write numbers up to a million in words and figures. 4. Solve problems on quantitative reasoning involving counting in thousands and millions. 5. Find the factors of numbers in each set. 6. Express numbers as product of prime factors.	Abacus, Number chart and table of factor chart.	Pupils to: 1. Count in thousands and millions 2. Read and write numbers in words and figures. 3. Solve problems on quantitative reasoning involving counting in thousands and millions 4. Identify prime numbers from 1 to 100 5. Express given numbers as product of prime factors.

MATHEMATIC CURRICULUM

THEME: NUMBERS AND NUMERATION
SUB-THEME: FRACTIONS

PRIMARY FIVE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
1. Fractions	Pupils should be able to: 1. change fraction to decimals and decimals to percentages and vice versa; 2. solve quantitative aptitude problems related to percentages; 3. state the relationship between fraction and ratio; 4. solve quantitative aptitude problem related to ratio.	i. Percentages ii. Ratio iii. Quantitative reasoning on ratio	1. Guides pupils to convert fractions to decimals and decimals to percentages and vice versa 2. Solve quantitative aptitude problems related to percentages 3. Guides pupils to state examples in which ratio is used. 4. Guides pupils in solving problems on sharing 5. Guides pupils to determine the ratio of 2 numbers. 6. Guides pupils to identify the relationship between ratio and fraction 7. Guides pupils to solve quantitative reasoning problems on ratio	1. Convert fraction to decimals and decimals to percentage and vice versa. 2. Solve quantitative aptitude problems related to percentages. 3. State examples in which ratio is used 4. Determine the ratio of two numbers 5. Identify the relationship between ratio and fraction. 6. Solve problems on ratios and fractions involving quantitative reasoning.	Fraction-Decimal conversion chart, Fraction-percentage chart, Decimal-percentage conversion chart, Percentage-Decimal conversion chart. Flash card	Pupils to: 1. Change fraction to decimal and decimal to percentage and vice versa 2. State the relationship between ratio and fraction 3. Solve quantitative reasoning problems on ratios 4. Find the ratios between two numbers

MATHEMATIC CURRICULUM

THEME: NUMBERS AND NUMERATION
SUB-THEME: FRACTIONS

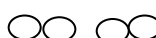
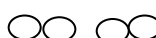
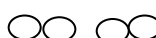
PRIMARY FIVE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
2. Multiplication	Pupils should be able to: 1. multiply a 3-digit number by a 3-digit number 2. solve quantitative aptitude problems on multiplication 3. apply "of" as multiplication when dealing with fractions of whole numbers.	i. Multiplication of 3-digit number by a 3-digit number ii. Quantitative reasoning multiplication iii. Using the operation "of" as multiplication in fraction.	1. Guides pupils to multiply a 3-digit number by a 3-digit number e.g. 437×132 . 2. Guides pupils to solve quantitative aptitude problems on multiplication e.g. 3. Guides pupils to apply the meaning of 'of' as multiplication such as: $\frac{1}{2}$ of 18 = 9 or $\frac{1}{2} \times 18 = 9$	1. Multiply a 3-digit number by a 3-digit number 2. Solve given problems on quantitative aptitude problems on multiplication 3. Bring their own bell to class 4. Apply the meaning of 'of' as multiplication in a fraction	1. Cardboard 2. Chart showing quantitative aptitude problems on multiplication 3. Orange ball etc	Pupils to: 1. Multiply a 3-digit number by a 3-digit number. 2. Solve quantitative aptitude problems involving multiplication of 3-digit numbers by 3-digit number. 3. Solve given problem on the operation of "of" as multiplication in fractions.

MATHEMATIC CURRICULUM

THEME: NUMBERS AND NUMERATION
SUB-THEME: FRACTIONS

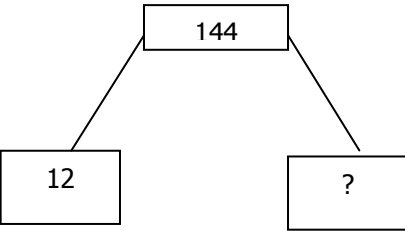
PRIMARY FIVE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE								
			TEACHER	PUPILS										
Multiplication contd.	Pupils should be able to: 4. multiply numbers by zero and one; 5. multiply decimals by whole numbers; 6. multiply decimal fractions by whole numbers.	iv. Multiplication of numbers by zero and one. iv. Multiplication of decimals by whole numbers iv. Multiplication of decimal fractions by whole numbers	4. Gives correct interpretation of zero and one as shown below. <table border="1"><tr><td></td><td>$2 \times 3 = 6$</td></tr><tr><td></td><td>$2 \times 2 = 4$</td></tr><tr><td></td><td>$2 \times 1 = 2$</td></tr><tr><td></td><td>$0 \times 0 = 0$</td></tr></table> 5. Guides the pupils to multiply given numbers by zero and one. 6. Guides pupils to solve the examples as follows (a) $2.86 \times 5 =$ or 2.86 x 5 14.30 (b) $1.27 \times 4 =$ or 1. 27 x 4 5.08 7. Guides pupils to solve problems on multiplication of decimal fractions by whole numbers.		$2 \times 3 = 6$		$2 \times 2 = 4$		$2 \times 1 = 2$		$0 \times 0 = 0$	5. Interpret the given problems 6. Solve problems on multiplication of numbers by 0 and 1 7. Multiply given decimals by whole numbers. 8. Carryout multiplication of decimal fraction by whole numbers.	Flip Chart Multiplication Charts	Pupils to: 4. Solves problems on multiplication by 0 and 1. 5. Multiply decimals by whole numbers. 6. Multiply decimal fractions by whole numbers.
	$2 \times 3 = 6$													
	$2 \times 2 = 4$													
	$2 \times 1 = 2$													
	$0 \times 0 = 0$													

MATHEMATIC CURRICULUM

THEME: NUMBERS AND NUMERATION
SUB-THEME: FRACTIONS

PRIMARY FIVE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
Multiplication contd.	Pupils should be able to: 7. calculate squares of whole numbers more than 50 and square roots of perfect squares greater than 400 8. solve quantitative aptitude problems involving squares of numbers more than 50 and square root of numbers greater than 400	V. Square of whole numbers more than 50 and square roots of perfect squares greater than 400	8. Guides pupils to find the square of given whole numbers more than 50 : 9. Guides pupils to find the square root of perfect square of whole number greater than 400 10. Guides pupils to solve quantitative aptitude problems e.g.  <pre> graph TD A[144] --- B[12] A --- C[?] </pre>	9. Find the square of a given whole number more than 50. 10. Find the square root of a perfect square of a whole number greater than 400. 11. Solve more quantitative aptitude problems on square roots and square of whole numbers.	1. Charts of whole numbers more than 50 and perfect square greater than 400 2. Charts on quantitative aptitude problems on square of numbers more than 50 and square root of numbers greater than 400	Pupils to: 7. Calculate the squares and square roots of given number more than 50 and greater than 400 respectively; 8. Solve quantitative aptitude problems on squares of numbers more than 50 and square roots of numbers greater than 400.

MATHEMATIC CURRICULUM

THEME: NUMBERS AND NUMERATION
SUB-THEME: FRACTIONS

PRIMARY FIVE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
3. Division	Pupils should be able to: 1. divide numbers by 10 and multiples of 10 up to 90; 2. solve quantitative aptitude problems involving division of number by 10 and multiples of 10 up to 900; 3. divide numbers by 100 and 200.	i. Division of numbers by 10, 20, ..., 90 ii. Quantitative reasoning on division iii. Division by 100 and 200	1. Guides pupils to determine how many group of 10, 20, 30 ... 90 are in a given number e.g. there are 3 groups of 10 in 30 Thus. $30 \div 3 = 3$ 2. Guides pupils to identify that in multiplying by 10, the decimal point is shifted once to the left to obtain the result of division 3. Guides pupils on how to solve quantitative problems on division $\begin{array}{r} 540 \\ 10 \overline{) } \end{array}$ $Q=54$ $\begin{array}{r} 540 \\ 20 \overline{) } \end{array}$ $S=$ 4. Guides pupils to divide numbers by 100 and 200 by shifting decimal point : i) twice to left when dividing by 100; ii) twice to the left when dividing by 200 and then divide by 2 e.g. $\frac{500}{100} = 5.00 = 5, \quad \frac{800}{200} = \frac{8.00}{2} = 4$	1. Determine the number of group to obtain 10,20,30 ...90 in a given number 2. Apply the rule of shifting decimal points once to the left to obtain result of divided numbers by 10 3. Solve given problems in quantitative aptitude on division 4. Carry out division of numbers by 100 and 200	1. Charts on division of number of 10 and multiples of 10 up to 90. 2. Charts containing worked problems involving division of number by 100 and 200	Pupils to: 1. Divide given numbers by 10 and multiples of 10. 2. Solve quantitative aptitude problems involving division of number by 10 and multiples of 10 up to 900 3. Solve given exercises on division by 100 and 200.

MATHEMATIC CURRICULUM

THEME: BASIC OPERATIONS SUB-THEME: BASIC OPERATIONS

PRIMARY FIVE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
1. Addition and subtraction	Pupils should be able to: 1. add and subtract numbers involving three or more digits. 2. add and subtract mixed fractions 3. solve quantitative aptitude problems involving addition and subtraction of fractions 4. add and subtract decimal fractions.	i. Addition and subtraction of whole numbers in involving three or more digits. ii. Addition and subtraction of mixed fractions and mixed numbers iii. Quantitative reasoning on addition and subtraction of fractions iv. Addition and Subtraction of decimal fractions	1. Guides pupils to add or subtract columns under unit first, tens and hundreds e.g. $ \begin{array}{r} \text{TH} \quad \text{H} \quad \text{T} \quad \text{U} \\ 5 \quad 6 \quad 7 \quad 4 \\ + 3 \quad 4 \quad 6 \quad 0 \\ \hline 2 \quad 5 \quad 7 \quad 0 \end{array} $ 2. Guides pupils to solve quantitative aptitude problems on addition and subtraction of fractions. 3. Guides pupils to add or subtract Numerator to get the sum. 4. Guides pupils in solving quantitative aptitude problems involving addition and subtraction of fractions. 5. Guides pupils to add and subtract decimal fractions.	1. Arrange counter into $ \begin{array}{cccc} & \text{TH} & \text{H} & \text{T} & \text{U} \end{array} $ 2. Carryout addition and subtraction of numbers 3. Carryout addition and subtraction of mixed fractions. 4. Use LCM method to add and subtract mixed fractions. 5. Solve quantitative aptitude problems involving addition and subtraction of fractions. 6. Add and subtract decimal fractions	Flash cards, Abacus etc Fraction chart Cardboard	Pupils to: 1. add and subtract numbers involving three or more digits 2. solve problems on quantitative aptitude involving addition and subtraction of fractions 3. add and subtract given fractions and mixed fractions 4.solve quantitative aptitude 5. add and subtract given decimal fractions

MATHEMATIC CURRICULUM

THEME: BASIC OPERATIONS
SUB-THEME: BASIC OPERATIONS

PRIMARY FIVE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
Addition and subtraction Contd.	Pupils should be able to: 4. solve quantitative aptitude problems involving division of numbers by 100 and 200 5. divide decimals by multiples of 10 up to 900 6. solves quantitative aptitude problems of decimals	iv. Quantitative reasoning on division of numbers by 100 and 200 v. Division of decimals by multiples of 10 up to 900 vi. Quantitative reasoning on decimals	1. Guides pupils to solve quantitative aptitude problems involving division by 100 and 200 e.g. $\begin{array}{r} 500 \\ 100 \overline{) 500} \\ \underline{100} \\ 400 \\ \underline{400} \\ 0 \end{array} = 5$ $\begin{array}{r} 200 \\ 20 \overline{) 400} \\ \underline{200} \\ 200 \\ \underline{200} \\ 0 \end{array} = 20$ 2. Guides pupils to divide decimals by multiples of 10 up to 90 by shifting decimal point e.g. (a) $\frac{32.5}{10} = 3.25$ (b) $\frac{710.4}{20} = \frac{71.04}{2} = 35.52$ 3. Guides pupils to solve quantitative aptitude problems such as: $\begin{array}{r} 10 \\ 30.8 \\ \underline{10} \\ 20.8 \\ \underline{20} \\ 0.8 \\ \underline{0.8} \\ 0 \end{array}$ $\begin{array}{r} 20 \\ 6.5 \\ \underline{20} \\ 0.5 \\ \underline{0.5} \\ 0 \end{array}$	1. Observe charts provides 2. Solves given problems in quantitative aptitude involving division by 100 and 200 3. Divide given decimals by multiples of 10 up to 90. 4. Solve given quantitative aptitude problems.	Quantitative aptitude charts with worked examples	Pupils to: 4. Solve quantitative aptitude problems involving division by 100 and 200 5. Solve given problems on division decimals by multiples of 10 6. Solves quantitative aptitude problems involving division of decimals by multiples of 10.

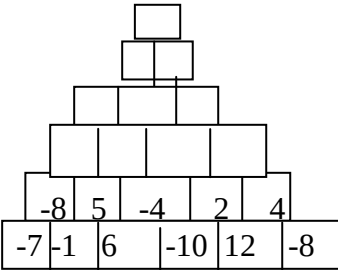
MATHEMATIC CURRICULUM

THEME: BASIC OPERATIONS
SUB-THEME: BASIC OPERATIONS

PRIMARY FIVE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
Addition and subtraction Contd.	Pupils should be able to: 7. divide decimals by 100 and 200 8. divide whole numbers by 2-digit numbers	vii. Division of decimals by 100 and 200 viii. Division of whole number by 2-digit numbers	1. Guides pupils to divide decimals by 100 and 200 e.g. $\frac{236.53}{100} = 2.3653$ 2. Guides pupils in the division of 2 whole numbers by 2 digit numbers as follows divide 357 by 21: $\begin{array}{r} 17 \\ 21 \overline{) 357} \\ \underline{21} \\ 147 \\ \underline{147} \\ 0 \end{array}$	1. Divide decimal fractions by 100 and 200. 2. Solve the division of whole numbers by 2-digit numbers.	Division charts of worked examples on division of decimals	Pupils to: 7. Divide given decimals by 100 and 200; 8. Solve problems on division by 2-digit number.

THEME: BASIC OPERATIONS
SUB-THEME: DERIVED OPERATIONS

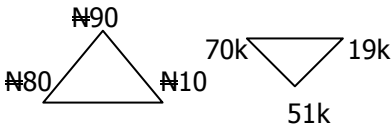
TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
1. Use of Number line in addition and subtraction	Pupils should be able to : 1 Add and subtract numbers using number line 2 Solve problems on quantitative aptitude involving addition and subtraction on the number line	1 Addition and subtraction of Positive and negative integers. 2 Quantitative reasoning problem involving number line.	1 Guides pupils to add and subtract numbers using number line. 2 Guides pupils to Solve problems on quantitative aptitude using number line Such as: Complete the pyramid 	1. Add and subtract numbers using the number line. 2 Solve problems on quantitative aptitude using the number line	Rule, Charts, etc	Pupils to: 1. add and subtract numbers using the number line. 2 solve problems on quantitative aptitude using number line.

THEME: ALGEBRAIC PROCESSES
SUB-THEME: ALGEBRAIC OPERATIONS

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
Open sentences	pupils should be able to: 1. find the missing number in open sentences. 2. use letters to represent boxes in open sentences 3. find the missing numbers that the letters represents. 4. interpret each box in a mathematical statement represent a letter that could be found 5. use letters to represent the missing numbers in quantitative aptitude problems and find their values.	Open sentences Quantitative reasoning on open sentences	1. Guides pupils to use letters to represent boxes e.g. $\square + 5 = 8$ $a + 5 = 8$ 2. Guides pupils to solve problems of the form $2t - 7 = 5$ 3. Guides pupils to solve quantitative aptitude problems on open sentences Such as.	1. Use letters to represent boxes in open sentences 2. Find the unknown 't' in the given statement 3. Solve quantitative aptitude problems on open sentences	flash cards and charts	Pupils to: 1. use letters to represent open sentences. 2. solve problems on open sentences 3. solve given quantitative aptitude problems on open sentences

THEME: MENSURATION AND GEOMETRY

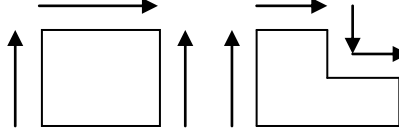
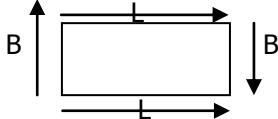
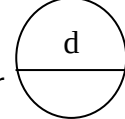
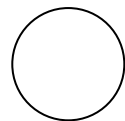
SUB-THEME: PRIMARY MEASURE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
1. Money	Pupils should be able to : - compare Nigerian units of money with pounds sterling, American dollars and some West African Countries; - solve problems on profit and loss, simple interest, commission, discount and transaction in the post offices, market etc; - solve quantitative reasoning problems on money.	- Nigerian Naira, pound Sterling, Dollars, Ghana Cedis and Pesewa, Sierra Leone's Leone and cents etc. - Money: social transactions, Home Banks, post Office, Market. - Quantitative reasoning on money transactions	- Guides pupils to view charts showing currency and its conversion rates of Naira to other currencies. - Guides pupils to explain that the demand (i.e. market force) for any currency will determine the conversion rates, hence fluctuation of conversion rates. - Guides the pupils to convert from one currency to another. - Guides pupils to carry out profit and loss, simple interest, commission, discount and the transactions in the offices, bank and market. - Guides pupils to solve quantitative reasoning problems on money e.g. 	- Carry out conversion of one currency to the other as contained in the chart. - Link rates of conversion to the purchasing power quoted in foreign currencies - Calculate profit and loss , simple interest, commission, discount and the transactions in the offices, banks and market - Solve quantitative reasoning problems involving money.	- Nigerian bank notes and coins, Foreign currencies, pictures and charts showing picture of currency rates. - Stamps, Nigerian bank notes and coins, models of money, shopping corner with goods and their carry prize tag - charts of solved examples on quantitative reasoning problems on money.	Pupils to: - Identify various currencies - convert one currency to another. - explain the implications of the rate of conversion on purchasing of the people - list their individual needs and ways of meeting them accordingly - calculate profit and loss simple interest and discount rates. etc - solve quantitative reasoning problems on money

MATHEMATIC CURRICULUM

THEME: MENSURATION AND GEOMETRY
SUB-THEME: PRIMARY MEASURE

PRIMARY FIVE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
2. Length	Pupils should be able to: - find the perimeter of regular shape such as square rectangle, trapezium and polygon; - find circumference of a circle when the radius is given.	- Perimeter of regular shapes e.g. square, rectangle trapezium and polygon - Circumference of a circle of given radius	- Leads pupils to discover that perimeter means total distance round a shape  - Guides pupils to use the formula $2(L+B)$ in calculating perimeter of square or rectangle as shown below  $P = L+B+L+B$ $= L+L+B+B$ $= 2L+2B$ $= 2(L+B)$ - Guide pupils to identify properties of circle such as: radius  diameter  Circumference 	- Find the perimeter of regular shapes - Find the perimeters of given objects. - Write the properties of a circle e.g. radius diameter and circumference in their exercise books - Find the circumference of circles with given radii	Chart containing regular shapes. Concrete objects that are circular in shapes, charts containing circle and its properties	Pupils to: - Find perimeter of regular shapes - state properties of circle - calculate the circumferences of circles with given radii.

MATHEMATIC CURRICULUM

THEME: MENSURATION AND GEOMETRY
SUB-THEME: PRIMARY MEASURE

PRIMARY FIVE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
Length Contd.	Pupils should be able to: 3. Establish the relationship between c and Π d and find the circumference	iii. Circumference of a circle with a given diameter	1. Guides pupils to find the circumference of a circle of given radius using $\text{circumference} = 2\Pi r$. 2. Guides pupils to find the circumference of a circle when the diameter is given. $C = \Pi d$.	1. Find the circumference of a circle when diameter is given.		Pupils to: 4. calculate circumference of a circle with given diameter.
3. Weight	Pupils should be able to: 1. solve word problems on weight 2. solve problems on quantitative aptitude involving weight	i. Word problems on weight involving kg and g ii. Quantitative reasoning on weight	1. Guides pupils to identify the weight of common goods in the environment and carry out addition, subtraction multiplication and division involving weight of goods. 2. Guides pupils to Solve quantitative aptitude problems related to weight.	1. Divide the total weight of pupils in the class by the total number of pupils in class. 2. Solve problems on quantitative aptitude involving weigh	Weighing scale, charts of weight of common goods: a bag of cement, a bag of ground-nut, a bag of rice, etc.	Pupils to: 1. Add the weights of given objects. 2. Solve problems on quantitative aptitude involving weights.
4. Time	Pupils should be able to: 1. calculate average speed of a moving object.	Average speed	1. Guides pupils to define average speed as. $\text{Average speed} = \text{distance} / \text{time}$ 2. Guides pupils to solve problems on average speed. 3. Guides pupils to solve word problems involving average speed.	1. Define average speed. 2. Find average speed in given problems. 3. Solve word problems involving average speed, etc.	Drawing of speedometer and cardboard showing some examples of average speeds	Pupils to find average speed of given word problems.
5. Temperature	Pupils should be able to: 1. compare degrees of hotness of various objects and areas (locations) in degree Celsius. 2. identify the usefulness of temperature to our daily life.	i. Familiarity with temperature of objects and towns in degrees (Celsius $^{\circ}\text{C}$)	1. Guides pupils to read thermometers to ascertain temperature of people, objects and locations.	1. Read thermometers to ascertain temperature of people, objects and locations.	Thermometer, data on meteorological information on some towns	Pupils to: 1. read temperature of given objects. 2. compare temperatures of of objects, town and locations.

THEME: MENSURATION AND GEOMETRY

SUB-THEME: SECONDARY MEASURE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	PUPILS		
1. Area	Pupils should be able to calculate the area of a right angle triangle.	Area of a right angled triangle.	1. Guides pupils to divide a rectangle into two halves along its diagonal to form two equal right angled triangles. 2. Guides pupils to derive the formula for area of right angled triangle i.e. $\frac{1}{2}$ of area of rectangle or $\frac{1}{2}$ (base x heights). 3. Guides pupils to calculate the area of the right angle triangle.	1. Derive and use formula to calculate the area of a right angled triangle	Charts, etc	Pupils to find the area of a given right angled triangle.
2. Volume	pupils should able to: 1. use cubes to find the volume of cuboids and cube; 2. use formula to find volume of cuboids; 3. identify the different between cubes and cuboids.	i. Volume of cuboids and cubes. ii. Volume of cuboids $V = L \times b \times h$ cubic unit	1. Guides pupils to count the number of cubes unit that makes up a cube or cuboids. 2. Guides pupils to find volume in unit cubes. 3. Guides pupils to measure length, breadth and height of cuboids and find the volume in cubic units.	1. Count number of unit cubes in cuboids. 2. Write the volume in cubic units. 3. Use unit cubes to build more cuboids. 4. Find volume of cuboid using the formular: $V = L \times b \times h$ cubic units	Unit cubes, etc Unit cube and cuboids.	Pupils to: 1. find volumes in cube units; 2. find given volume of cuboids using $L \times b \times h$

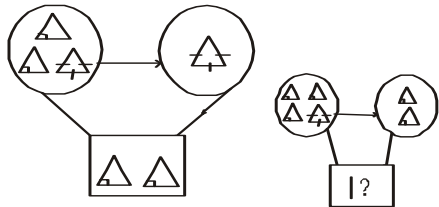
MATHEMATIC CURRICULUM

THEME: MENSURATION AND GEOMETRY
SUB-THEME: SECONDARY MEASURE

PRIMARY FIVE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	PUPILS		
3. Capacity	Pupils should be able to : 1. find the relationship between litres and cubic centimeters. 2. identify the use of litre as a unit of capacity and the established relationship between litre and cm^3	Litres as cm^3 : 1 litre = 1000 cm^3	1. Guides pupils to compare the volume of the open cube and that of the litre container. 2. Guides pupils to identify litre as a unit of capacity and the relationship between litre and cm^3	1. Compare capacity of the container and the cube of dimension $10\text{cm} \times 10\text{cm} \times 10\text{cm}$ 2. Identify that 1 litre = 1000 cm^3 3. Establish the relationship between litre and cm^3	Litre, capacity container Cube of dimension $10\text{cm} \times 10\text{cm} \times 10\text{cm}$	Pupils to: 1. construct their own cube of dimension $10\text{cm} \times 10\text{cm} \times 10\text{cm}$. 2. compare the liquid content of their model with that of a litre container and comment.
4. Structure of Earth	Pupils should be able to 1. describe shape of earth. 2. compare volume of a sphere and cuboid	Shape of earth and volume of sphere	1. Guides pupils to describe the shape of earth usually globe 2. Puts the globe in an open box of dimension $L \times b \times h$ 3. Guides pupils to identify that the volume of globe is less than that of enclosing cuboid	1. Describe earth as a spherical object 2. Calculate volume of the cuboids. 3. Shows that the volume of the globe is less than the volume of the cuboid enclosing it.	Globe, cardboard, box, oranges, etc.	Pupils to say which is bigger, volume of sphere or volume of the cuboid that encloses it.

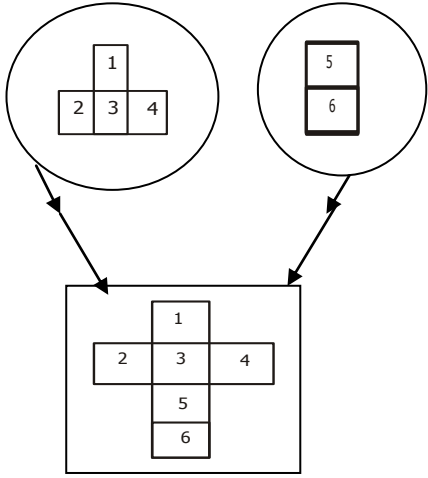
THEME: MENSURATION AND GEOMETRY
SUB-THEME: SHAPES

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
1. Plane shapes	Pupils should be able to: 1. identify parallel and perpendicular lines. 2. solve quantitative aptitude problems on plane shapes. 3. state some properties of triangles including equilateral, isosceles and right angle triangle. 4. solve some quantitative aptitude problems involving triangles	i. Parallel and perpendicular lines. ii. Quantitative aptitude on parallel and perpendicular lines. iii. Triangles: • Equilateral • Isosceles • Right-angled. iv. Quantitative reasoning on triangles	1. guides pupils to explain parallel and perpendicular lines using edges of the board. 2. guides pupils to identify parallel and perpendicular lines using objects in the classroom. 3. guides pupils to use symbols for parallel lines and perpendicular lines 4. guides pupils to solve quantitative aptitude problems using symbols for parallel lines and perpendicular lines 5. guides pupils to discover the features of equilateral triangle, isosceles triangle and right angled triangle. 6. leads pupils to solve some quantitative aptitude problems e.g. 	1. Identify parallel and perpendicular lines in selected objects 2. Label parallel lines and perpendicular lines 3. Solve quantitative aptitude problems with symbols for parallel and perpendicular lines 4. State the features of equilateral triangle, isosceles triangle and right angled triangle. 5. Solve some quantitative aptitude problems on triangle	2 and 3 dimensional shapes Models of: Equilateral, Isosceles, Right angled triangles	Pupils to: 1. Identify parallel and perpendicular lines. 2. Solve quantitative aptitude problems on parallel and perpendicular lines. 3. State two properties each of an equilateral, isosceles and right-angled triangles 4. Solve given quantitative aptitude triangle problems.

MATHEMATIC CURRICULUM

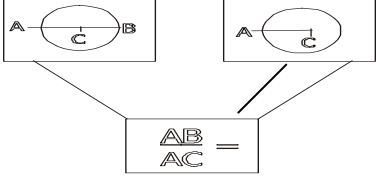
THEME: MENSURATION AND GEOMETRY SUB-THEME: SHAPES

PRIMARY FIVE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
2. 3-dimensional shape	Pupils should be able to: 1. State properties of 3 dimensional shapes such as cubes, cuboids, pyramids, etc 2. solve quantitative aptitude problems related to 3 dimensional shapes such as cubes, cuboids and pyramid etc.	i. Cube, cuboid, pyramid and a square base, and triangular base. ii. Quantitative reasoning related to 3-dimensional shapes	1. Guides pupils to identify properties of 3 dimensional shapes such as cube, cuboid, and cylinder etc. 2. Guides pupils to solve quantitative aptitude problems related to cube, cuboid and pyramid such as 	1. State properties of cubes, cuboids and pyramids. 2. Solve quantitative aptitude problems related to cube, cuboid and pyramid.	Graph paper, cardboard sheets, models of 3 dimensional shapes, etc.	Pupils to: 1. Give the properties of 3-dimensional shapes 2. Solve given quantitative aptitude problems relating to three dimensional shapes.

MATHEMATIC CURRICULUM

THEME: MENSURATION AND GEOMETRY
SUB-THEME: SHAPES

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
3. Circle	Pupils should be able to: - identify - radius - diameter - circumference of a circle - solve quantitative aptitude problems on circle. - Identify and determine a radius on the diameter of the circumference of a circle	- Circle - Radius - Diameter - Circumference - Quantitative reasoning on circle	- Guides pupils to measure the distance from the center of any point on the circumference and vice versa. - Guides pupils to measure distance round the circle to determine the circumference. - Guides pupils to solve some quantitative aptitude problems on circles. - Guides pupils to identify the relationship between radius and diameter of the circles 	- Measure the distance from the centre to any point on the circumference - State the relationship between radius and diameter - Determine the distance round the circle and determine the circumference of the circle. - Solve some quantitative aptitude problems on circle.	strips of cards, pencil, pin, tray, sand of a given circle etc	Pupils to: - Draw circles of different radius and obtain diameter for the circle - Obtain the Circumference of the circles with different radius - Solve some problems on quantitative aptitude relating to circles.

MATHEMATIC CURRICULUM

PRIMARY FIVE

THEME: EVERYDAY STATISTICS

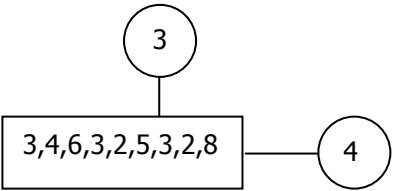
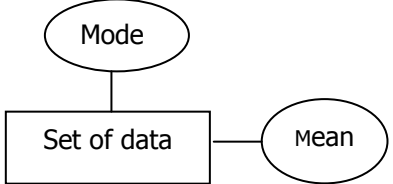
SUB-THEME: DATA COLLECTION AND PRESENTATION

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
1. Data presentation	Pupils should be able to: 1. prepare a tally of data 2. draw bar graphs and pictograms of information collected locally	i. Further work on pictograms and bar graphs ii. Use tally and tables	1. Guides pupils to Select data on test results of pupils in the class (full mark 25) or select data from the mathematical game or activity designed by the teacher e.g. from a card or other games. 2. Guides pupils to use tally to represent the information. 3. guides pupils to Identify and represent information using data from events on daily life activities. 4. Present data generated by tally in tabular form. 5. Guides pupils to represent data on pictogram. 6. Guides pupils to represent data on bar graphs.	1. Select and Records pupils score 2. Use tally to represent the information 3. Present the data in tabular form 4. Represent the information using pictogram 5. Represent the information on the bar graph	1. Data on test results. 2. data on weather 3. data on election 4. biological data 5. teachers game or activity etc.	Pupils to: 1. Prepare and present a tally of data and put it in a tabular form 2. Draw pictogram and bar graph of a given data 3. prepare a tally of a set of data. 4. Construct pictogram and bar graph of a given data.
2. Measures of central tendency	The pupils should be able: 1.find the mode of given data 2.identify the mode as applicable in daily life activities 3.Calculate the mean of a given data 4.Identify mean of a set of data in daily life activities 5.Solve quantitative aptitude problems on mode and mean of data	iii. Mode of a given data	1. Leads the pupils to get data from their environment and ask them to calculate the mode. 2. Guides pupils data and asks them to prepare a tally of a data and find the mode.	1. Prepare tally of data and record the mode. 2. Carry out experiment to get data from the environment and find the mode.	Data chart, etc	Pupils to: 1. Find the mode of a given set of data.

MATHEMATIC CURRICULUM

THEME: EVERYDAY STATISTICS
SUB-THEME: DATA COLLECTION AND PRESENTATION

PRIMARY FIVE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
Measures of central tendency contd.	Pupils should be able to: 3. calculate the mean of given data. 4. appreciate the concept of mean of a set of data in daily activities	vi. Mean of data v. Quantitative reasoning	3. Guides pupils to calculate the mean from a given data 4. Guides pupils to calculate mean from data gathered from their environment e.g. average number of students in each arm of the class. 5. leads the pupils to solves quantitative reasoning problems such as  Which is equivalent? 	3. Calculates the mean from the given data 4. Solve quantitative reasoning problems on mode and mean	Data chart.	Pupils to: 2. calculate mean of given data 3. calculate the mean from data gotten from the environment daily activities 4. solve quantitative reasoning problems on mode and mean of given data

MATHEMATIC CURRICULUM

PRIMARY FIVE

THEME: EVERYDAY STATISTICS
SUB-THEME: CHANCE AND EVENTS

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
1. Tossing coins and throwing of die	Pupils should be able to: 1. Record in data from , experiments on coin tossing and dice throwing 2. Identify various chance events in their daily life activities	iii. Tossing of coins and throwing of dice. iv. Other chance events	1. Guides pupils to toss a coin 20 times and record the number of times a head appears and the number of times a tail appears. 2. Guides pupils to prepare a tally for their results 3. Guides pupils to throw a dice 24 times and record occurrence of 1,2,3,4,5,6 4. Guides pupils to prepare tally for their results 5. Guides the pupils to identify various chance events in their daily life activities.	1. Toss the coin 20 times and records the number of times a head appears and number of times the tail appears. 2. Prepare a tally of heads and tails (20 tosses) 3. Throw the die 24 times and records the number of times of occurrence of 1,2,3,4,5 and 6 4. Prepare tally from the results 5. Identify various chance events in their daily life activities.	Coin, die.	Pupils to performs experiments as directed by the teacher, and record the result by tallying.

MATHEMATIC CURRICULUM

THEME: NUMBER AND NUMERATION
SUB THEME: WHOLE NUMBERS

PRIMARY SIX

505 THREE-FOUR WHOLE NUMBERS																																																												
TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE																																																						
			TEACHER	PUPILS																																																								
1. Whole Numbers	Pupils should be able to: 1. count in millions and billions; 2. write and read up to one million; 3. solve problems involving quantitative reasoning; 4. give the place value and the value of a digit in a given whole number; 5. identify numbers in place value.	i. Counting in millions and billions. ii. Writing and reading up to one million. iii. Quantitative reasoning. iv. Value and place value of whole numbers.	1. Guides the pupils to revise the previous work in counting up to counting in millions. 2. Guides pupils to count in millions up to nine hundred millions. 3. Guides pupils to read up to one millions as shown below: <table><tr><th>Numeral</th><th>word form</th></tr><tr><td>1,000,000</td><td>one million</td></tr><tr><td>10,000,000</td><td>ten million</td></tr><tr><td>50,000,000</td><td>fifty million</td></tr><tr><td>100,000,000</td><td>one hundred million</td></tr><tr><td>500,000,000</td><td>five hundred million</td></tr><tr><td>1000,000,000</td><td>1 billion</td></tr></table> 4. Devises other practices or methods to write and read up to one billion. 5. Guides pupils to solve problems in quantitative reasoning. 6. Guides the pupils in revising the previous work on place value and value of a digit. 7. Guides pupils to find the place value and value of numbers e.g. The number 347840 has the value and place value of each digit as follows: <table><tr><td>3</td><td>=</td><td>300,000</td><td></td></tr><tr><td>4</td><td>=</td><td>40,000</td><td></td></tr><tr><td>7</td><td>=</td><td>7,000</td><td>}values</td></tr><tr><td>8</td><td>=</td><td>800</td><td></td></tr><tr><td>4</td><td>=</td><td>40</td><td></td></tr><tr><td>0</td><td>=</td><td>0</td><td></td></tr><tr><td>Hth</td><td>tth</td><td>th</td><td>h</td><td>t</td><td>u</td><td>} place</td></tr><tr><td>3</td><td></td><td>4</td><td></td><td>7</td><td>8</td><td>4</td><td>0</td><td>} value</td></tr></table>	Numeral	word form	1,000,000	one million	10,000,000	ten million	50,000,000	fifty million	100,000,000	one hundred million	500,000,000	five hundred million	1000,000,000	1 billion	3	=	300,000		4	=	40,000		7	=	7,000	}values	8	=	800		4	=	40		0	=	0		Hth	tth	th	h	t	u	} place	3		4		7	8	4	0	} value	1. Revise counting up to one millions. 2. Count in millions up to nine hundred million. 3. Write numbers up to one million. 4. Devise other practices or methods to write and read up to one billion. 5. Solve problems in quantitative reasoning. 6. Give the value and place value of each of the digits in the given number.	Abacus and Charts of numbers, etc	Pupils to: 1. count in millions up to a specified number; 2. write up to one million; 3. solve given problems on quantitative reasoning; 4. give the place value of each of the digits in a given number.
Numeral	word form																																																											
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MATHEMATIC CURRICULUM

THEME: NUMBER AND NUMERATION
SUB THEME: WHOLE NUMBERS

PRIMARY SIX

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
Whole Numbers (Contd)	Pupils should be able to: 1. solve problems on quantitative reasoning with value and place value; 2. give the value and place value for a digit in a decimal fractions; 3. solve quantitative aptitude problems related to place value; 4. find LC M of 2-digits whole numbers.	i. Quantitative reasoning. ii. Value and place value of decimal fractions vii. LC M of 2-digits whole numbers.	1. Guides pupils to solve problems in quantitative reasoning on value and place value. 2. Guides pupils in revising previous work on decimal fractions. 3. Guides pupils to find the place value and value of decimal fractions e.g. The number 634.578 has the value and place value of each digit as follows: $6 = 600$ $3 = 30$ $4 = 4$ $5 = 5/10 \text{ or } 0.5$ }value $7 = 7/100 \text{ or } 0.07$ $8 = 8.1000 \text{ or } 0.008$ H t u t h } place 6 3 4 5 7 8 }value 4. Guides pupils to solve quantitative aptitude problems related to place value 5. Guides pupils to revise previous work on LCM of 1-digit numbers 6. Guides pupils to write out multiples of numbers say 10 and 12. 7. Guides pupils to list the common multiples of 10 and 12 as 60, 120, 240, ... and choose the least common multiple (LCM) i.e. 60. 8. Guides pupils to find LCM by factor method as: $10 = 2 \times 5$ $12 = 2 \times 2 \times 3$ $LCM = 2 \times 2 \times 3 \times 5 = 60$	1. Solve problems in qualitative reasoning. 2. Revise previous work on decimal fractions 3. Give the value and place value of the decimal fractions. 4. Solve quantitative aptitude problems on place value 5. Revise the previous work on LCM of 1-digit numbers. 6. List multiples of 10 and 12. 7. Find the LCM of 10 and 12. 8. Find the LCM of 10 and 12 by factor method. 9. Compare the two methods of finding LCM.	Chart of numbers, Abacus Relevant charts, Charts of factors and multiples of numbers.	Pupils to: 1. solve problems on quantitative 2. give the value and place value of each digit in a given decimal fractions reasoning 3. solve quantitative aptitude problems on place value 4. find the multiples of given 2- digit numbers 5. find the LCM of given 2-digit numbers by ▪ factor method ▪ multiples method

MATHEMATIC CURRICULUM

THEME: NUMBER AND NUMERATION SUB THEME: WHOLE NUMBERS

PRIMARY SIX

HCF OF TWO 2-DIGIT WHOLE NUMBERS												
TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE						
			TEACHER	PUPILS								
Whole Numbers (Contd)	Pupils should be able to: 1. find HCF of 2-digits whole numbers; 2. solve quantitative aptitude problem on LCM and HCF.	i. HCF of 2-digits whole numbers ii. Quantitative reasoning	1. Guides pupils to revise previous work on HCF of 1-digit numbers 2. Guides pupils to list the factors of 2-digits numbers say 28 and 30 i.e. Factors of 28 are 1,2,4,7,14,28 Factors of 30 are 1,2,3,5,6,15,30. 3. Guides pupils to list the common factors of 28 and 30 as 1 and 2, then choose the Highest Common Factor (HCF) i.e. 2. 4. Guides pupils to find HCF by factor method as: 28 = 1 x 2 x 2 x 7 30 = 1 x 2 x 3 x 5 Common factors = 1 and 2 HCF = 1 x 2 = 2 6. Guides pupils to solve quantitative aptitude problem on LCM and HCF.	1. Revise the previous work on HCF of 1-digit number. 2. using common factor to find HCF e.g. <table><tr><td>2</td><td>28</td><td>30</td></tr><tr><td></td><td>14</td><td>15</td></tr></table> HCF of 28 and 30 3. List factors of 28 and 30 4. Find the HCF of 28 and 30 by factor method. 5. Solve quantitative aptitude problem on LCM and HCF	2	28	30		14	15	Charts of factors of numbers	Pupils to: 1. find the factors of given whole numbers; 2. find the HCF of given whole numbers; 3. solve some quantitative aptitude problems involving HCF; 4. solve quantitative aptitude problem on LCM and HCF.
2	28	30										
	14	15										

MATHEMATIC CURRICULUM

THEME: NUMBER AND NUMERATION
SUB THEME: WHOLE NUMBERS

PRIMARY SIX

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	PUPILS		
2. Demography	Pupils should be able to: 1. read, write and compare populations of small and big cities; 2. read and compare population of HIV prevalence in different countries; 3. use counting in thousand and million in population studies.	i. Population: ▪ families ▪ classes ▪ towns and cities etc. ii. Reading and comparing of populations	1. Guides pupils to define population. 2. Guides pupils to give the populations of: ▪ Their houses ▪ classes ▪ Town etc. 3. Guides pupils to read and write populations of some cities e.g. Lagos, Kano, Aba in the country using map of Nigeria. 4. Guides pupils to read and compare population of HIV prevalence in different countries of the world 5. Guides pupils to use counting in thousands and millions in population studies	1. Define population. 2. Give population of their houses, classes, towns etc. 3. Read and write populations of some cities e.g. Lagos, Kano, Aba in the country using the map of Nigeria. 4. Read and compare population of HIV prevalence in different countries in the world. 5. Use counting in thousands and millions in population studies.	Demography map of Nigeria Video of populations World Atlas and some published information from WHO on HIV/AIDS etc Flip charts	Pupils to: 1. define population; 2. read and write the population of big cities in Nigeria; 3. read and compare population; 4. read and compare population of HIV prevalence in some given countries of the world; 5. give population of countries in Africa.

MATHEMATIC CURRICULUM

THEME: NUMBER AND NUMERATION SUB THEME: FRACTIONS

PRIMARY SIX

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	PUPILS		
1. Fractions	Pupils should be able to: 1. order fractions; 2. solve problem on quantitative reasoning; 3. express decimals as fractions and vice versa.	i. ordering of fractions ii. Quantitative reasoning iii. Decimals	1. Guides pupils to order fractions by first bringing the fractions to have the same denominator and consider the denominators e.g. In the fractions, $\frac{1}{2}$ and $\frac{3}{5}$ $\frac{1}{2} = \frac{1 \times 5}{2 \times 5} = \frac{5}{10}$ $\frac{3}{5} = \frac{3 \times 2}{5 \times 2} = \frac{6}{10}$ $5 < 6$ then $\frac{5}{10} < \frac{6}{10}$ Hence $\frac{1}{2}$ is smaller than $\frac{3}{5}$. 2. Guides pupils to solve problems on quantitative aptitude e.g. Insert the appropriate symbol $<$ or $>$ in the diagram below $1\frac{1}{2} \quad \square \quad \frac{5}{2}$ 3. Guides pupils to express decimals as fractions and vice versa e.g. $0.75 = \frac{75}{100} = \frac{3}{4}$ L $2\frac{1}{2} = \frac{2 \times 2 + 1}{2} = \frac{5}{2}$ $= \frac{5 \times 5}{5 \times 2} = \frac{25}{10} = 2.5$	1. Order fractions by first bringing the fractions to have the same denominator and consider the denominators. 2. Solve problems on quantitative reasoning involving fractions. 3. Express decimals as fractions and vice versa e.g.	Charts, flash charts Flip charts of decimal fractions	Pupils to: 1. order given fractions; 2. solve problems on quantitative aptitude involving fractions; 3. express given decimals as fractions and vice versa.

MATHEMATIC CURRICULUM

THEME: NUMBER AND NUMERATION SUB THEME: Fractions

PRIMARY SIX

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
2. Ratios and Proportions	Pupils should be able to: 1. solve problems on ratio; 2. apply ratio to everyday life; 3. solve quantitative reasoning problem involving ratio; 4. solve problems in direct proportion; 5. solve problems on quantitative reasoning involving direct proportion; 6. solve problems on inverse proportions; 7. identify some daily life activities that are inversely related; 8. solve problems on quantitative reasoning in inverse proportions	i. Ratios ii. Direct proportion iii. Quantitative reasoning v. Inverse proportions vi. Quantitative reasoning	1. Guides pupils to revise previous work done on ratios 2. Guides pupils to solve problems on population e.g. in a city of 10,000 people 100 of them are HIV positive. Find the ratio of those infected to the total population. Total population 100,000 Infected population = 100 The ratio of infected to total population is 100:10,000 or 1:100 or 1/100. 3. Guides pupils to apply ratio to everyday life. 4. Guides pupils to solve quantitative problems related to ratios. 5. Guides pupils to solve direct proportion problems e.g. A man saves money everyday. If he saves ₦30 each day, how much can be saved in 4 days? Saving (₦) day 30 1 60 2 90 3 120 4 He saves ₦120 in 4 days. 6. Guides pupils to note that his saving is in direct proportion to the number of days. 7. Guides pupils to solve problems on quantitative reasoning involving direct proportion. 8. Guides pupils to solve problems on inverse proportions. 9. Guides pupils to identify some activities that are inversely related. 9. Guides pupils to solve problems on quantitative reasoning in inverse proportions.	1. Revise the previous work on ratios 2. Solve problems on ratios 3. Apply the application of ratios in everyday life. 4. Solve quantitative reasoning problem related to ratios. 5. Solve examples on direct proportion. 6. Note that the saving is in direct proportion to the number of days. 7. Solve quantitative aptitude problems involving direct proportion 8. Solve given problems in inverse proportion. 9. Identify some daily life activities that are inversely related. 10. Solve problems on quantitative reasoning in inverse proportion.	Source for related charts Flash charts Inverse proportion charts Inverse population charts	Pupils to: 1. solve given problems on ratios; 2. solve some quantitative reasoning problem on ratios; 3. solve given problems on direct proportions; 4. solve problem on quantitative aptitude involving direct proportion; 5. solve given problems in inverse proportions; 6. solve given problems in quantitative reasoning in inverse proportions.

MATHEMATIC CURRICULUM

THEME: NUMBER AND NUMERATION SUB THEME: Fractions

PRIMARY SIX

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
3. Ratio of family size and resources	Pupils should be able to: 1. find ratio of family size and resources; 2. correctly use ratio in relating proportion of resources to family size.	Ratio of family size and resource	1. Guides pupils to solve problems on ratio of family size and resource e.g. ▪ Tunde's family size is 12 and he earns ₦36,000 monthly. Find the ratio of his family size to his salary ratio is 12:36000 or 1:3000 → ₦3000 3. Guides pupils to correctly use ratio in relating proportion of resources to family size.	1. Solve given problems by finding the ratio of family size to salary and state how much each member will get. 2. Correctly use ratio in relating proportion of resources to family size.	A poster depicting two families one smaller and happy and the other larger and unhappy Chart with population figures	Pupils to: 1. calculate the ratio of given family size to their earning; 2. give four examples of everyday activities that demand correct application of ratio.
4. Ratio of two populations	Pupils should be able to express two populations in given ratio.	Expressing two populations in ratio form	1.Guides pupils to express two populations in a ratio e.g. ▪ Two families ▪ Two towns ▪ Two countries.	1. Find the ratio of two given populations.	Charts with population figures	Pupils to find the ratio of given populations.

MATHEMATIC CURRICULUM

THEME: NUMBER AND NUMERATION
SUB THEME: FRACTIONS

PRIMARY SIX

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
5. Ratio of prevalence of HIV/AIDS between two sexes, two states	Pupils should be able to: - express the ratio of prevalence of HIV/AIDS between two sexes in a town or Country; - correctly express the ratios in monitoring the trend of infection of HIV/AIDS in different sexes and among states - solve quantitative aptitude problems on ratio of HIV/AIDS prevalence.	- Expressing the ratio of prevalence of HIV/AIDS between two sexes, two states etc. - Quantitative reasoning	- Guides pupils to express the prevalence of HIV/AIDS by sex in ratio using the chart. - Guides pupils to correctly express and apply ratio in everyday activities such as HIV/AIDS for the purpose of having a clear picture of the trend. - Guides pupils to solve quantitative aptitude problems on ratio of HIV/AIDS prevalence.	- Express the prevalence of HIV/AIDS of sexes in ratio. - Correctly express and apply ratio in everyday activities such as HIV/AIDS for the purpose of having a clear picture of the trend - Solve quantitative aptitude problems on ratio of HIV/AIDS prevalence.	Data Chart of HIV/AIDS prevalence Data chart on HIV/AIDS	Pupils to: - find the given ratio of prevalence of HIV/AIDS by sex of a population; - give five examples of issues of HIV/AIDS that require correct expression of ratios; - solve quantitative aptitude problems on ratio of given HIV/AIDS prevalence.

MATHEMATIC CURRICULUM

THEME: NUMBER AND NUMERATION SUB THEME: FRACTIONS

PRIMARY SIX

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
6. Percentages	1. Pupils should be able to: 1. express one number as a percentage of another; 2. calculate numbers expressed as a percentage of another; 3. express one population as a percentage of another population; 4. solve problems on increase percent.	i. Expressing a number as a percentage of another number ii. Percentage increase	1. Guides pupils to express numbers as a percentage of another e.g. finds the percentage of boys and girls in a class. 2. Guides pupils to calculate numbers expressed as a percentage of another. 3. Guides pupils to calculate percentage increase as $\text{Percentage increase} = \frac{\text{increase}}{\text{initial value}} \times 100$ 4. Guides pupils to solve problems on percentage increase e.g. In 2000 the number of Nigerians infected with HIV/AIDS was 100,000 and in 2004 the number was 250,000 calculate the percentage increase.	1. Express numbers as a percentage of another e.g. find the percentage of boys and girls in a class. 2. Calculate numbers expressed as a percentage of another 1. Solve problems on percentage increase. 2. Find the percentage increase in HIV/AIDS infection of the problems given.	Source for correct data on HIV/AIDS Data chart on HIV/AIDS	Pupils to: 1. express given numbers as a percentage of another; 2. calculate given numbers expressed as a percentage of another; 3. Solve problems in percentage increase involving HIV/AIDS infection and other related problems.

MATHEMATIC CURRICULUM

THEME: NUMBER AND NUMERATION
SUB THEME: FRACTIONS

PRIMARY SIX

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
Percentage contd.	5. correctly calculate percentage increase such as in population growth; 6. solve problems on percentage decrease; 7. solve problems on quantitative aptitude on percentage increase and decrease.	iii. Percentage decrease iv. Quantitative reasoning	1. Guides pupils to correctly calculate percentage increase in everyday activities such as in population growth. 2. Guides pupils to give more examples of everyday activities where percentage increase is needed. 3. Guides pupils to calculate percentage decrease as follow: $\text{Percentage} = \frac{\text{decrease}}{\text{initial value}} \times \frac{100}{1}$ E.g. Calculate the decrease in infection of HIV/AIDS since 2002 given following figure of infected persons 2002 - 2,500 2005 - 1,800 $\text{Percentage} = \frac{700}{2500} \times \frac{100}{1} = 28\%$ Decrease 2500 1 4. Guides pupils to Solve problems in quantitative aptitude on percentage increase or decrease e.g. 2002 2500 2005 1800 28% 2005 1800 2006 1700 ?	1. Calculate percentage increase of a given population growth. 2. Gives examples of everyday activities that require percentage increase. 3. Calculate percentage decrease of a given population growth. 4. Solves quantitative aptitude on percentage decrease	Charts with population figures Charts with population figures Charts with population figures	Pupils to: 4. calculate percentage increase of a given population growth; 5. calculate percentage decrease of a given population growth; 6. solves quantitative aptitude on percentage increase and decrease.

MATHEMATIC CURRICULUM

THEME: BASIC OPERATIONS SUB THEME: BASIC OPERATIONS

PRIMARY SIX

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVELUATION GUIDE
			TEACHER	PUPILS		
1. Addition and Subtraction	Pupils should be able to: 1. add any set of numbers; 2. solve problems on subtraction of whole numbers; 3. solve word problems involving addition and subtraction of whole numbers.	i. Addition of whole numbers. ii. Subtraction of numbers. iii. Addition and subtraction (word problems)	1. Guides pupils to solve problems involving addition of whole number. 2. Guides pupils to solve problems on subtraction of whole numbers. 3. Guides pupils to solve word problem in addition and subtraction of whole	1. Solve problems involving addition of whole numbers. 2. Solve problems on subtraction of whole numbers. 3. Solve word problems involving addition and subtraction.	Flash cards of numbers Flash cards of numbers Flash cards of numbers	Pupils to: 1. solve given problems on addition of whole numbers; 2. solve given problems on subtraction of whole numbers; 3. solve given word problems involving addition and subtraction.

MATHEMATIC CURRICULUM

THEME: BASIC OPERATIONS SUB THEME: BASIC OPERATION

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
2. Multiplication	Pupils should be able to: - multiply a 3-digit number by a 3-digit number. - Apply multiplication to the daily life activities - solve problems on quantitative aptitude on multiplication. - multiply decimal by decimal (to one decimal place). - multiply fractions by fraction - calculate square of numbers up to 500. - calculate the square roots of perfect squares.	- Multiplication by 3-digit numbers - Multiplication of decimal by decimal - Multiplication of fractions by fraction - Squares of numbers up to 500 - Square roots of perfect square 4,9,25,100,900	- Drills pupils on multiplication. - Guides pupils to solve multiplication problems on 3 by 3- digits numbers e.g. $\begin{array}{r} 572 \\ \times 321 \\ \hline 572 \\ 1144 \\ 1716 \\ \hline 183612 \end{array}$ - Solves problems on quantitative aptitude on multiplication. - Guides pupils to solve problems on the multiplication of decimals by a decimal. - Guides pupils to solve problems on multiplication of fractions by fraction e.g. $\frac{3}{4} \times \frac{8}{9} = \frac{3 \times 8}{4 \times 9} = \frac{2}{3}$ $1\frac{1}{3} \times 2\frac{1}{4} = \frac{4}{3} \times \frac{9}{4} = 3.$ - Guides pupils to solve problems on square of numbers e.g. Square of 5 is $5^2 = 5 \times 5 = 25$ Square of 15 = $15^2 = 15 \times 15$ - Guides pupils to solve problems on square roots using factorization method.	- Solve problems on multiplication on 3 of 3 -digits numbers. - Apply multiplication to daily life activities. - Solve problems on quantitative aptitude on multiplication. - Solve problems on the multiplication of decimals by a decimal. - Solve problems in multiplication of fraction by fractions. - Solve problems on square of numbers. - Solving problems on square roots using factorization method.	Sum cards containing multiplication facts and charts of solved problems Charts of decimal numbers Charts of fractions Perfect square chart Perfect square charts	Pupils to: - solve given problems involving multiplication by 3-digit number; - solve problems on quantitative aptitude involving multiplication. - multiply given decimals by decimal; - solve exercise on multiplication of fractions by fraction; - calculate squares of given numbers up to 500; - find the square roots of given perfect squares.

MATHEMATIC CURRICULUM

THEME: BASIC OPERATIONS SUB THEME: BASIC OPERATIONS

PRIMARY SIX

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
3. Division	Pupils should be able to: 1. divide whole number and decimal by 2-digit and 3-digit numbers.	Division of whole numbers and decimal by 2-digit and 3-digit numbers.	1. Guides the pupils to divide whole number by 2-digit numbers. 2. Guides pupils to solve problems on division of whole numbers and decimal by 2-digit and 3-digit numbers 3. Guides pupils to solve quantitative problems on division of whole numbers and decimals by 2-digit and 3-digit numbers.	1. Solve problems on division of whole numbers and decimals by 2-digit and 3-digit numbers. 2. Solve quantitative problems on whole numbers and decimals by 2-digit and 3-digit numbers.	Flash cards and charts.	Pupils to: 1. solve given problems on divisions by 2-digit and 3-digit number; 2. solve problem on quantitative aptitude involving division.
4. Order of operations	Pupils should be able to: 1. use basics operations in the right order. 2. solve quantitative aptitude problems involving BODMAS	i. Order of operations (BODMAS) ii. quantitative reasoning	1. Explain BODMAS as showing the order of operation i.e. B (bracket) is done first O (of) is done next to bracket D (Division) follows M (multiplication) is next to division A (Addition) is next to multiplication S (Subtraction) is taken last. 2. Guides pupils to solve problems using BODMAS 3. Guides the pupils to solve problems in quantitative aptitude involving BODMAS .	1. Note and write the meaning of BODMAS . 2. Solve problems using order of OPERATIONS.	Flip charts of operations involving BODMAS	Pupils to : 1. solve problem using BODMAS; 2. solve quantitative aptitude exercise involving BODMAS.

MATHEMATIC CURRICULUM

THEME: BASIC OPERATIONS
SUB THEME: DERIVED OPERATIONS

PRIMARY SIX

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
1. Indices (power)	Pupils should be able to: 1. write numbers in index form; 2. solve problems involving powers (indices); 3. solve problem on quantitative reasoning involving indices.	i. Numbers in index form (power not exceeding 5). ii. Indices	1. Introduce powers by using the idea of square e.g. $2 \times 2 = 2^2$, $3 \times 3 = 3^2$ 1. Guides the pupils to express numbers in index form e.g. express 48 in index form $48 = 2 \times 2 \times 2 \times 2 \times 3 = 2^4 \times 3$ 2. Guides pupils to multiply numbers using the index form e.g. $20 \times 63 = (2^2 \times 5) \times (3^2 \times 7)$ 3. Guides pupils to solve problems on quantitative reasoning in index form.	1. Solve problems in index form. 1. Express numbers in index form 2. multiply numbers using index form. 3. Solve problems on quantitative aptitude related to indices		Pupils to: 1. write given squares in index form; 2. express given numbers in index form; 3. multiply given numbers using index form; 4. solve problem on quantitative reasoning related to indices.
2. Ratio and percentage	Pupils should be able to: 1. solve problems on ratios; 2. express a number as a percentage of another number; 3. solve problems on profit and loss percent; 4. apply the profit and loss percent to business; 5. solve problem on quantitative aptitude on ratio, percentage, profit and loss.	i. Ratio and percentage ii. Expressing a number as a percentage of another number iii. Profit and loss percent	1. Guides pupils to solve problem on ratio, e.g. the ratio of boys to girls in a class is 3:2. If there are 35 pupils in the class, how many are boys? Total ratio = $3 + 2 = 5$ No. of boys = $2/5 \times 35 = 21$. There are 21 boys in the class. 2. Guides pupils to express a number as a percentage of another e.g. what percentage of 10 is 5. $= 5/10 \times 100/1 = 50\%$ 3. Guides pupils to calculate the profit and loss percents in a given problem. 4. Guides pupils to solve quantitative aptitude problems on ratio, percentage, profit and loss.	1. Calculate ratios of given numbers. 2. Express given numbers as percentages of others. 3. Calculate profit and loss percents of given problems. 4. Solve quantitative aptitude problems on ratio, percentage, profit and loss.	Charts containing ratios and percentages Charts containing numbers Profit and loss charts	Pupils to: 1. calculate ratios of given numbers; 2. express a number as a percentage of another; 3. calculate profit and loss percents of given problems; 4. solve quantitative aptitude problems on ratio percentage profit and loss.

MATHEMATIC CURRICULUM

THEME: ALGEBRAIC PROCESSES

SUB THEME: OPEN SENTENCES

PRIMARY SIX

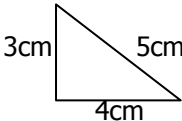
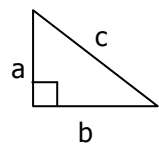
TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
1. Open sentences	People should be able to 1. Solve problems expressed as open sentence. 2. Interpret words in open sentences and solve them. 3. solve related problems on quantitative aptitude.	Open sentences	1. Guides pupils to interpret and solve word problems e.g. if 60 oranges are divided equally in 3 baskets how many are in one basket? 3. Guides pupils to solve quantitative aptitude related problems involving three or more arithmetic operations in a sample	1. Interpret word problems into open sentences and solve them. 3. Solve quantitative aptitude problems involving three or more arithmetic operations in a sample.	1. Open sentence chart	Pupils to: 1. interpret word problems into open sentences and solve them; 2. Pupils to solve quantitative aptitude problems involving at least three arithmetic operations in a sample.

MATHEMATIC CURRICULUM

THEME: MEASUREMENT SUB THEME: PRIMARY MEASURES

PRIMARY SIX

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
1. Money (a) Population and Economics consequences	Pupils should be able to: 1. solve problems on taxes and rates; 2. solve problem involving buying and selling of share and dividends; 3. solve problem on quantitative aptitude based on taxes and rate as well as selling of share and dividends.	i. Taxes, rates, rents, buying and selling of shares and dividends.	1. Guides pupils to give reasons for some levies. 2. Guides pupils in the transaction of buying and selling of shares. 3. Guides pupils to calculate taxes, rates, and shares dividends. 4. Guides pupils to solve problem on quantitative aptitude based on taxes and rates as well as selling of shares and dividends.	1. List items on which taxes and rates could be taken. 2. Calculates taxes, rates and shares dividends. 3. Solve problems on quantitative aptitude based on taxes and rates as well as selling and share and dividends	Chart on market pricing index	Pupils to: 1. list various forms of taxes; 2. calculate taxes, rates and shares dividends; 3. solve quantitative aptitude problems.
(b) Conversion of currencies	1. use a calculator to convert from one currency to another; 2. solve problems on quantitative aptitude based on converting from one currency to another.	ii. Use of calculator in conversion of currency.	1. Guides pupils to use calculator for the conversion of one currency to another. 2. Guides pupils to solve problem on quantitative aptitude based on converting from one currency to another.	1. Use calculator to convert currencies. 2. Solve problems on quantitative aptitude bases on converting from one currency to another.	Calculator	Pupils to: 1. use calculator to convert currencies; 2. solve quantitative aptitude problems.

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
2. Length	Pupils should be able to: 1. use Pythagoras rule to find the unknown length of a right angled triangle. 2. solve many quantitative aptitude problems involving Pythagoras rule	Pythagoras rule	1. Guides pupils to draw a right-angled triangle with sides 3cm, 4cm and 5cm. i.e.  2. Guides pupils to identify the hypotenuse, opposite and the adjacent sides of the right angled triangle. 3. Guides the pupils to use Pythagoras rule to calculate the unknown side i.e. $a^2 + b^2 = c^2$ - where a and b are the adjacent and opposite sides of the triangle and c the hypotenuse side  4. Guides pupils to solve quantitative aptitude problems involving Pythagoras rule.	1. Draw the right angle triangle. 2. Use Pythagoras theory $a^2 + b^2 = c^2$ to calculate the unknown side 3. Solve quantitative aptitude problems on Pythagoras rule.	Chart on Pythagoras theorem	Pupils to: 1. find unknown length of a given right angled triangle using the Pythagoras formula; 2. solve given quantitative aptitude problems on Pythagoras rule.

MATHEMATIC CURRICULUM

THEME: MEASUREMENT SUB THEME: PRIMARY MEASURES

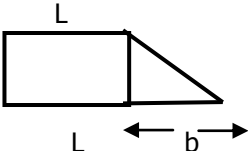
PRIMARY SIX

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
3. Weight	Pupils should be able to: 1. express the same weight in different unit gram, kilogram, tone; 2. Solve word problems on weights.	Word problem on weights	1. Guides pupils to obtain measures of their weight or other objects using weighing scales. 2. Guides pupils to express the weights in different units 3. Guides pupils to solve word problems on weight.	1. Convert weights to different units. 2. Solve word problems on weights.	Samples of different objects, weighing scales and spring balance.	Pupils to: 1. convert weights expressed in tones to kilogram and vice versa; 2. solve word problem involving weights.
4. Time	Pupils should be able to: 1. tell time in seconds and minutes; 2. Solve quantitative aptitude problem on time; 3. Read timetable of journeys especially by trains and aero planes; 4. Use time to plan daily activities.	i. Timing in minutes and seconds. ii. Reading time-table of journeys	1. Guides pupils to perform practical activities to determine time on task e.g. walking, eating, etc. 2. Guides pupils to solve quantitative aptitude problem on time. 3. Guides pupils to use these time table to estimate the time it takes a train or an aeroplane to get to a specific destination. 4. Guides pupils to prepare timetable for their daily or weekly activities.	1. Measure time on task in seconds and minutes. 2. Solve quantitative aptitude problems involving timing. 1. Use time table to estimate the time it will take a train and aeroplane to get to a designated destination. 2. Use timetable to plan for a day or a week.	1. Second pendulum, stop-watch and clock. 2. School Time-table or calendar, list schedule of flight.	Pupils to: 1. measure time on task; 2. solve quantitative aptitude problems involving timing; 3. plan for a week or two weeks; 4. obtain time table of train and estimate the time it will take the train to get to specific destination.

MATHEMATIC CURRICULUM

**THEME: MEASUREMENT
SUB THEME: SECONDARY MEASURES**

PRIMARY SIX

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
1. Perimeter	Pupils should be able to discover that difference rectangle with same area have different perimeters	Perimeter of a rectangle	1. Guides pupils to draw two rectangles of dimensions 20cm by 10cm and 25cm by 8cm. 2. Guides pupils to prove that they have equal area but different perimeters	1. Find the perimeters of the two rectangles. 2. Compare their results.	Cardboard showing rectangle of equal areas but different perimeters.	Pupils to: 1. calculate perimeters of rectangles with equal areas but with different lengths and breadths.
2. Area	Pupils should be able to: 1. calculate the areas of figure which can be divided into rectangles and or triangles; 2. calculate land areas in hectares.	i. Areas of trapezium and other figures. ii. Land areas	1. Guides pupils to draw and break the trapezium into two parts (rectangle and triangle) e.g.  The area of the trapezium is (Area of Rectangle) + (Area of Triangle) i.e. $(L \times H) + (h \times b / 2)$. 2. Guides pupils to calculate land areas expressing the units in hectares.	1. Calculate the area of the given trapezium. 2. Find areas in m^2 and convert to hectares	Nail board, rubber bands, and cut out shape from cardboards Land plan	Pupils to: 1. find the areas of figures which can be divided into rectangles and triangles; 2. calculate the areas of the given shapes or figures. 3. Calculate given areas in square metres and convert to hectares.

MATHEMATIC CURRICULUM

THEME: MEASUREMENT

SUB THEME: SECONDARY MEASURE

PRIMARY SIX

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	PUPILS		
3. Volume	Pupils should be able to: 1. calculate volume of prisms, cylinders and spheres; 2. solve some quantitative aptitude problem on volume of prism, cylinder and sphere.	Volume of prism, cylinders and sphere	1. Guides pupils to calculate volumes of prism, cylinders and spheres. 2. Guides pupils to solve some quantitative aptitude problem on volume of prism cylinder and spheres	1. Calculate volumes of prism, cylinders and spheres. 2. Solve quantitative aptitude problems on volume	Balls, cylindrical tins, triangular and prisms	Pupils to calculate volumes of given triangular prism, cylinders and spheres.
4. Capacity	Pupils should be able to: 1. To use standard units; 2. solve word problems involving litres.	Word problem on litres	1. Guides pupils to work in groups to compare the capacities of the containers given to them e.g. different sizes of graduated containers. 2. Guides pupils solve word problems on capacity e.g. ▪ A drum contains 1500 litres of petrol. How long will it last if a driver takes 50 litres a day?	1. Calculate capacities of given containers. 2. Solve word problems on capacity.	1. Standard bottles or container of Ragolis or Swan water, buckets of different sizes.	Pupils to: 1. find the capacities of common containers in their respective home. Record the results and bring them to the class for discussion; 2. solve word problems on capacities.

MATHEMATIC CURRICULUM

THEME: MEASUREMENT SUB THEME: SECONDARY MEASURES

PRIMARY SIX

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
5. Speed	1. Pupils should be able to: 1. calculate average speed; 2. solve quantitative aptitude problems on time and speed.	Average speed	1. Guides pupils to calculate average speed as $\frac{\text{distance covered}}{\text{time}}$ 2. Guides pupils to solve quantitative aptitude problems related to time and speed.	1. Calculate average speed using $\frac{\text{distance covered}}{\text{time}}$ 2. Solve quantitative aptitude problems related to time and speed.	Clock face, stop watch, railway time-table, bus time-table, air time-table	Pupils to: 1. calculate speed when time and distance are given.
6. Athletics time	Pupils should be able to estimate time to complete races.	Standard time for races	1. Guides the pupils to give record times for 100m, 400m, 800m by men and women from different competitions for different years.	Estimate time taken for different races.	Time of race completion charts	Pupils to estimate time a boy or girl will take to run 100m.

MATHEMATIC CURRICULUM

THEME: MENSURATION AND GEOMETRY SUB THEME: SHAPES

PRIMARY SIX

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	PUPILS		
1. Plane Figures	Pupils should be able to identify the basic properties of plane figures such as: rectangle and a square	1. Basic properties of plane figures.	- Guides pupils to identify properties of plane figures such as square and rectangular shapes. - Guides pupils to identify properties such as equal sides, parallel sides, right angles, line of symmetry, diagonals of square and rectangle.	Identify properties of plane figures such as square and rectangular shapes.	Rectangular and square shapes, metre rule/tape, and protractors	Pupils to state the properties of a square and a rectangle.
2. Angles	Pupils should be able to: - measure angles in degrees; - measure angles in a plane.	1. Measurement of angles	- Guides pupils in the correct placement of protractors to measure a given angle in degree. - Guides pupils to indicate the edges, vertices and faces of given three dimensional shapes. - Guides pupils to measure the size of angle, indicate lines that are parallel and perpendicular in each of the given shape.	- Use protractor and ruler to measure correctly the given angle. - Identify the number of edges vertices and faces for different shapes. - Use protractors to determine the sizes of angles in a given shape. - Identify lines that are parallel and perpendicular in three dimensional shapes.	Mathematical set, chalkboard and protractor Protractors, net of polygons, adhesive tapes, graph sheets.	Pupils to: - measure given angles from diagram in their text books, work books etc. - draw various plane shapes and measure resulting angles to the nearest degree; - identify the edges, vertices and faces of given three dimensional shapes; - measure the sizes of angle in a two dimensional shapes.

MATHEMATIC CURRICULUM

THEME: MENSURATION AND GEOMETRY SUB THEME: SHAPES

PRIMARY SIX

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
3. Height and distances	Pupils should be able to measure heights of human beings building, trees and distance	Measurement of height and distances	1. Guides pupils to obtain readings of their heights in metre using the wall metre rule. 2. Guides pupils to read measurements of distance and heights using click wheel.	1. Measure their heights (without shoes) using the wall metre rule. 2. Use metre rule to obtain measures of the dimensions of the classroom, flower lawns etc. 3. Use tapes to measure longer distances.	Metre rule, wall metre rule, click wheel, steel measuring tapes	Pupils to measure and obtain the dimensions of classroom, house, play grounds, halls etc as project * Note that the units for these measurements are in metres and kilometers
4. Polygons	Pupils should be able to: 1. solve more problems on two dimensional and three dimensional shapes.	1. Two dimensional figures not exceeding octagon and three dimensional shapes	1. Guides the pupils to recall the names and features of given shapes 2. Guides pupils to identify line of symmetry, number of sides, number of angles, etc.	1. Identify a given plane shapes in terms of its name and essential features e.g. square, rectangle, triangle rhombus, parallelograms etc. 2. Using paper cutouts to determine the number of lines of symmetry, number of angles and number of sides.	Charts of two and three dimensional figures.	Pupils to Complete a chart to indicate - Number of lines of symmetry - Number of sides - Number of angles contained in given shape.
5. Scale Drawing	Pupils should be able to: 1. draw plans according to a given scale; 2. apply and use scale drawing in converting lengths and distances of objects in his /her environment to any scale.	(i) Scale drawing	1. Guides pupils in converting given lengths and distances to their corresponding scale values. 2. Takes out pupils to measure the dimensions of farms, garden, playground etc and reduce it to a given scale.	1. Convert given lengths and distance using a defined scale 2. Draw the floor plan of their classroom, school farms, gardens and play grounds using a given scale. 3. Determine actual values/dimensions of figures drawn to scale.	Different scale, rules, tapes, protractors	Pupils to: 1. convert given lengths and distances to a given scale; 2. draw to scale school farms, play grounds etc with their dimension given; 3. convert scale plans to their real dimensions.

MATHEMATIC CURRICULUM

THEME: EVERYDAY STATISTICS

SUB THEME: DATA COLLECTION AND PRESENTATION

PRIMARY SIX

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	PUPILS		
1. Population	Pupils should be able to: 1. interpret pictograms and bar graphs. 2. use pictogram and bar graphs in representing population of people or data.	Pictograms and bar graphs	1.Guides pupils to present data such as: men - 20,000 women - 30,000 boys - 40,000 girls - 36,000 on a bar graph.	1. Present the population data of men, women, boys and girls in the bar graph. 2. Represent a given population data on a bar graph or pictogram.	Population data chart.	Pupils to Prepare a given population data on pictogram or bar graph.
2. Measures of Central Tendency	Pupils should be able to: 1. find the mode of data; 2. calculate the mean of given data.	i. mode ii. mean	1. Guides pupils to collect and summarize data and indicate the mode. 2. Guides pupils to organize a given data and calculates the mean.	1. Organize a given data and indicate the mode. 2. Organize a given data and calculate the mean.	Data chart Set of data	Pupils to: 1. organize a given data and find the mode; 2.organize a given data and calculate the mean.